



Forside

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Abstract

This bachelor project explores the practical application of KartAi technologies to streamline the building application process in collaboration with Kristiansand Municipality. The project builds upon a previous Proof of Concept by integrating AI agents into a functional prototype aimed at reducing complexity and improving the submission quality of one-step building applications. Emphasizing a user-centered approach, the project focused on improving accessibility for residents and minimizing the workload for municipal case workers through guided form flows, integrated chatbot assistance, and real-time spatial and document validation tools. Using the agile Scrum methodology, the group iteratively developed, tested, and refined the solution across 17 sprints, whilst engaging the stakeholder throughout. The resulting prototype demonstrates how AI technologies can be applied to real municipal workflows, helping applicants submit more complete and accurate applications. While the system remains a prototype, it offers a solid foundation for further development within the broader KartAi initiative, with documented recommendations for expanding functionality, including a potential interface for case workers and enhancements on chatbot performance and data integration.

A walkthrough of the product can be viewed [here](#).

The Github repo with the application can be found [here](#).

Preface

This bachelor's thesis marks the end of our studies in IT and Information Systems at the University of Agder and was written as part of the course IS-304 *Bachelor's Thesis in Information Systems*. Over the course of our education, we have gained valuable knowledge and skills which we have applied and further developed through this project.

The thesis was conducted from January to May 2025 as a collaboration between three students and Kristiansand Municipality within the KartAi initiative, combining academic learning with practical experience.

We thank Kristiansand Municipality for their support and contributions. A special thanks to Eva for her direction and consistent support, Dagfinn for his valuable input, Mari for sharing her experience, Sébastien for his energy and fresh perspectives, and Rune for his reliable help and insights. Your involvement has been key to the project's success.

We are also grateful to our university supervisor, Katherine Brown, for her guidance, availability, and expertise throughout the process.

Finally, thanks to our course coordinators, Hallgeir Nielsen and Geir Inge Hausvik, for their support and for providing a solid academic framework.

We hope this thesis will contribute to ongoing improvements in the digitalization of the building application process and the Norwegian public sector's digital transformation.

Kristiansand

May 16th, 2025

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Statement of Usage of Artificial Intelligence

In this project, AI tools were strategically employed to support both technical development and report writing. GitHub Copilot was used within the coding environment to provide context-aware code suggestions, which streamlined development and increased overall productivity. ChatGPT contributed by offering technical explanations and guidance for resolving complex issues during implementation.

Furthermore, AI was utilized to improve the linguistic quality of the report. ChatGPT assisted in rephrasing and editing some sentences of the text to enhance clarity, coherence, and flow. This ensured that the report met academic standards in both content and presentation.

Although AI tools can raise concerns about over-reliance and potential inaccuracies (IEEE, 2023), we addressed these risks by verifying AI-generated outputs against reliable sources and ensuring that decisions were not based solely on AI responses. This approach maintained the accuracy and reliability of the project.

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Central Concepts

To navigate effectively through the report, it is essential to understand the key concepts in the project “Utstillingsvindu 2.0 – Practical Application of KartAi Technology”. This subchapter introduces and explains the key concepts, to ensure a uniform understanding of technical terminology.

Agile project management: Agile project management is a methodology for managing software development projects through iterative cycles. It emphasizes adaptability, continuous delivery, and the integration of customer feedback throughout the development process. Each iteration, or sprint, aims to produce a functional product increment, allowing for regular reassessment and improvement (Atlassian, n.d.,-a).

Artificial Intelligence (AI): Artificial Intelligence (AI) refers to systems or applications designed to simulate human intelligence, including capabilities such as learning, reasoning, and decision-making (Stryker & Kavlakogul, 2024).

AI Agent: An AI agent is a software system that autonomously performs tasks or pursues goals on behalf of users. It leverages AI to reason, plan, learn from experience, and adapt its behavior over time.

Application Programming Interface (API): An API is a set of rules that allows software applications to communicate and share data, features, or functionality with one another (Goodwin, 2024).

Building Application (Byggesøknad): A building application is a formal submission comprising forms and documentation that detail a proposed construction project, submitted to municipal authorities to obtain the necessary building permit (Forbrukertorget, n.d.,).

Embedding: An embedding is a mathematical representation of data, such as text, images, or audio, used by machine learning models to capture the underlying meaning or features of that data for tasks like classification or semantic search (Cloudflare, n.d.,).

Large Language Models (LLMs): Large Language Models (LLMs) are advanced AI models trained on vast datasets, enabling them to understand and generate natural language and other types of content across a variety of tasks (IBM, 2023).

One-step application (Ettrinssøknad): A one-step application is the primary form used to apply for a building permit when the project is fully designed and requires formal responsibility assignments for the work (Byggesøknaden, n.d.,).

Proof of Concept (PoC): A Proof of Concept is an early-stage process used to determine whether a proposed project or solution is feasible. It helps evaluate potential before significant time and resources are invested (Atlassian, n.d.,-b).

Scrum: Scrum is an agile framework that organizes teamwork into roles, events, and artifacts to deliver value through iterative development (Scrum.org, n.d.).

Sprint: A Sprint is a time-boxed period, usually up to one month, in which a Scrum team develops a usable product increment (Scrum Guides, 2020).

User Interface (UI): UI refers to the visual and interactive elements of a digital product that users engage with, such as buttons, menus, and layouts (Figma, n.d.).

User Experience (UX): UX is the overall experience and satisfaction a user has when interacting with a product, focusing on ease of use and functionality (Finn & Downie, 2025).

Virtual Machine (VM): A virtual machine is a software-based emulation of a physical computer, running its own operating system and applications while sharing the physical resources of a host system (Microsoft, n.d.).

1 Introduction

This report is the end result of the journey to attain the Bachelor's degree in IT and Information Systems from the University of Agder (UiA). The report has been written as a part of the course IS-304 *Bachelor's Thesis in Information Systems*, where the main purpose is to apply acquired knowledge and skills accumulated throughout the degree by delving into a real setting project related to an information system (University of Agder, n.d.-a). The University of Agder states that:

“Students are given the opportunity to plan, estimate, execute, test, follow up and document the project. The development is done by a student group who are given the opportunity to work with quality in process and product, by using methods and techniques that promote these elements.” (University of Agder, n.d.-a)

This encompasses the key learning outcomes of the course and the degree, laying the foundation of the report and the project it concerns. Since the course is project based, collaboration between the student group and an external company is necessary. The student group, Group 13, or better known as Atlas within the project, has established a collaboration with the municipality of Kristiansand within the greater sphere of the research project KartAi (KartAi, n.d.). Overall, the assigned project aims to further establish and develop the existing Proof of Concept (PoC) which demonstrates how AI agents can assist applicants and case workers in the Norwegian building application process.

The report begins by presenting the project's context, criteria, and objectives, followed by an overview of the collaboration with Kristiansand Municipality and the student group's internal

structure. It then outlines the chosen project management approach, including the use of the Scrum methodology, risk management, quality assurance, and time estimation. The methodology chapter follows, describing the overall development strategy, data collection methods, stakeholder involvement, and analysis techniques used to guide the project. The report then moves into an in-depth analysis of user needs and system requirements, followed by design choices and architectural decisions. It continues with a description of the technologies and tools used, and a chronological account of the project execution through weekly sprints. The system is then tested and evaluated, before the final product is presented along with its functionality and potential for further development. The report concludes with a reflection on the process and outcomes, and a summary of the overall results in relation to the original goals.

1.1 Background

KartAi is a collaborative initiative aimed at enhancing municipal property and building case management through automated, data-driven methods and Artificial Intelligence (AI), while also prioritizing proactive user and citizen engagement (KartAi, n.d.). Led by Kristiansand Municipality in partnership with Norkart AS, Kartverket, and the University of Agder, the project spans from 2021 to 2027 (KartAi, n.d.). Its primary objectives include developing scalable, standardized solutions for seamless data integration across municipalities, advancing machine learning models to support decision-making, automating case processing for greater efficiency and reliability, and ensuring the ethical application of AI to foster socially sustainable and effective workflows in Norway.

The KartAi project is highly relevant due to the project's goal to help automate and streamline case processing in construction cases (KartAi, n.d.). The latest statistics from 2023 show that the number of building applications processed by the Norwegian municipalities fell by 7% from 2022 to 2023, and by a total of 11% since 2019. Despite this decline, the average processing time for one-step applications has remained unchanged at 19 weeks (Statistics Norway, 2024). This suggests that municipalities are spending the same amount of time or potentially more on each application, indicating that the average processing time per case has increased. The statistics highlight a growing inefficiency within the system and reinforce the need for digital tools that can streamline workflows and reduce the administrative burden for both applicants and case workers.

The group connected with Kristiansand Municipality during a matchmaking event between student groups and companies called RefreshIT, hosted by the University of Agder in October 2024. The municipality proposed a project within KartAi that aimed at improving the building application process to enable the public to submit applications more efficiently. They had a prior project with a student group from NTNU, who created a PoC that demonstrated how AI agents could support both applicants and case workers in the building application process (Aarø et al., 2024). However, the municipality wanted the PoC to be further developed through the practical use of the AI agents in the application process, rather than a showcase of the technology itself. This formed the foundation of the project, which

focuses on how KartAi's AI agents can be practically implemented in the building application process.

1.2 Project Criteria

This section defines the scope, problem, research focus, and objectives that guided the project's development. It also outlines the initial boundaries that shaped what was realistically achievable within the timeframe and resource constraints.

1.2.1 Scope and Direction

At the start of the project, Kristiansand Municipality intentionally left the scope open to provide the group flexibility in shaping the project's direction in close collaboration with them. As mentioned, the municipality had previously worked with a student group from NTNU, and their project would in later iterations move from a technological demonstration and towards practical integration of AI models within a user-centered prototype.

In the early stages, the group adopted an exploratory approach, engaging in stakeholder discussions, analyzing the existing PoC, and evaluating various directions the project could take. The group considered multiple perspectives, including legal, ethical, and comparative approaches. Ultimately, the group chose a user-centered focus, emphasizing usability and accessibility, primarily from the applicants' perspective. While the needs of municipal case workers were also considered, the group's initial research indicated that the biggest challenges lie in the applicant's experience, particularly in navigating the complexity of the application process. Due to time and capacity limitations, the group chose to prioritize addressing applicant-facing issues, in particular one-step applications, while still acknowledging the importance of case workers in the overall workflow.

1.2.2 Problem Statement

A key challenge in the current building application process is that many applications are incomplete or contain incorrect information. This not only causes delays but also increases the workload for municipal case workers. Through early research and stakeholder discussions, it became evident that the complexity of the building application process poses a significant barrier for residents. The problem can be summarized as follows:

“Despite a decline in the number of building applications, processing times have increased. The building application process remains highly complex, making it difficult for residents to understand the necessary requirements, navigate the process, and determine where and how to apply. As a result, many applications are submitted with missing or incorrect information, creating additional workload for case workers. This inefficiency leads to delays, unnecessary obstacles, and extended processing times. Therefore, there is a need for a solution that facilitates the submission of complete and accurate applications. This would then streamline

the overall case processing workflow and improve efficiency for both applicants and case workers.”

This statement forms the foundation for the research questions and project objectives that follow.

1.2.3 Research questions

To ensure a structured investigation, the following research questions were formulated:

1. How can the building application process be digitalized and made more efficient using AI and data-driven methods, combined with proactive citizen engagement?
2. How can residents intuitively submit complete digital applications?
3. How can case workers receive complete and accurate applications to reduce processing time?

1.2.4 Objectives

This project addresses the problem statement presented earlier, which highlights delays in the building application process due to incomplete and/or incorrect submissions. While the research questions addressed this issue at a broader, exploratory level, the objectives below represent the concrete goals and deliverables that guided the group’s work in collaboration with Kristiansand Municipality.

The project was guided by the following objectives:

- Enhance the existing PoC by refining AI-driven features that support both applicants and case workers.
- Improve the user experience through front-end development and UX/UI design.
- Assess the functionality of AI agents in providing regulatory guidance and identifying errors in building applications.
- Deliver a functional prototype that demonstrates the practical application of AI in a building application process.
- Provide documentation and recommendations to support further development of the KartAi project.

1.2.5 Initial Boundaries and Limitations

The project began with flexibility in direction, but from the outset, it operated within certain constraints to remain achievable within the given timeframe and available resources. One of the main challenges throughout the project was managing this open-ended scope. Since the final direction depended on what the group chose to focus on in collaboration with Kristiansand Municipality, the range of possibilities was initially very broad. As a result, the

group had to continually evaluate and narrow down the scope to ensure the project remained focused, realistic, and manageable.

These boundaries were mutually agreed upon by the student group and the stakeholder to ensure the project's feasibility while still aiming for a meaningful and functional outcome. The main limitations included:

- **Limited AI Agent Development:** The project did not involve building entirely new AI agents. Instead, the focus was on integrating and visualizing existing models provided by the KartAi project. However, during development, it became evident that one of the AI chatbots required refinement to improve its accuracy and relevance. Although this task fell outside the original plan, the group decided, after discussions with the stakeholder, that addressing this issue was necessary to ensure usability. This refinement process is discussed in more detail in section 6.2 Processes and Components and 9.2 Manual Testing of Chatbot.
- **Dependency on Ongoing AI Development:** A key challenge was that many of the AI models developed under the broader KartAi project were still in progress during the timeframe of this project. This created uncertainty around their stability, functionality, and documentation, making it more difficult for the group to map, test, and incorporate them effectively into the prototype. As a result, certain features had to be adjusted or simplified based on the availability and maturity of the AI components.
- **Prototype over Full-Scale Deployment:** Given time and resource constraints, the project did not aim to deliver a fully integrated or production-ready system. Instead, the outcome was a high-fidelity prototype that extended the existing PoC and demonstrated key functionality. This prototype was developed to showcase the practical use of AI models in the building application process and to serve as a foundation for further development within the KartAi project.
- **Focus on User Experience:** The group prioritized usability, accessibility, and a user-friendly interface, particularly for applicants. Consequently, less emphasis was placed on optimizing back-end systems or advancing AI logic beyond what was necessary for the prototype.

A critical reflection on these limitations and their impact on the project outcome is provided in section 11 Reflection.

1.3 Client Overview

Kristiansand Municipality is the primary stakeholder in this bachelor project. The project is part of the broader KartAi initiative, which involves multiple partners contributing different types of expertise (KartAi, n.d.). However, amongst these, Kristiansand Municipality had the most direct influence on the project. While Norkart AS and Kartverket offered technical and data-related contributions (Kartverket, n.d.; Norkart AS, n.d.), and the University of Agder acted as a research and development partner (University of Agder, n.d.-b), the municipality served as the product owner. They defined the problem space, provided access to real-world

data and workflows, and evaluated usability based on practical needs from a public administration perspective.

Because Kristiansand Municipality was the primary stakeholder, the final product prioritized their needs over those of other, more peripheral partners. The group therefore focused less on data analysis and developing new AI agents, and more on addressing concrete user challenges and integrating under-development AI components into a functional prototype. The goal was to demonstrate how these technologies could support a more efficient and user-friendly building application process.

Throughout the project, the group worked closely with five representatives from Kristiansand Municipality, each contributing valuable insights and expertise:

1. **Product Owner:** Served as the main point of contact and ensured alignment with KartAi's goals from the municipality's perspective.
2. **Technology Advisor:** Provided insight into the municipality's existing digital systems and the underlying KartAi technologies.
3. **Building Case Worker:** Shared domain-specific knowledge related to planning and building application processes.
4. **AI Advisor:** Offered guidance on how to apply and adapt AI technologies within the project.
5. **Technical Specialist:** Contributed programming expertise and technical support. *This individual joined the project approximately two months after its initiation, offering valuable input in the later stages of development.*

1.4 The Group

The project group consists of three students. Two had previously worked together, while the third had collaborated with only one of them on earlier projects. This mix of familiarity and new dynamics helped create a balanced and productive working environment. Over the course of the project, the group adapted to each member's working style and effectively leveraged individual skill sets to support one another and reach shared goals.

To maintain a structured and efficient workflow, the group divided responsibilities based on each member's strengths, interests, and prior experience. While everyone had a primary role, the group embraced a flexible and collaborative approach that allowed members to contribute across different areas when needed. This structure promoted both specialization and a shared understanding of the project's broader scope.

An overview of each member's primary responsibilities and additional contribution is presented in Table 1. Moreover, a reflection of the individual member's contribution to the project can be found in Appendix 2.

Group Member	Primary Role	Key Responsibilities
Anna	Group Leader & UX Designer	Project coordination, communication with stakeholder,

		meeting logistics, UX design, and report writing.
Maria	Front-end Developer	Development of responsive UI, component integration, quality assurance, and design implementation.
Nicoleta	Full Stack Developer (<i>Back-end focus</i>)	Database design, API integration, chatbot refinement, and AI agent/model integration.

Table 1 - Overview of Group Members Responsibilities

2 Project Management

Effective project management was essential to ensure that work progresses in a structured goal-oriented manner while balancing available resources, time constraints, and quality expectations. In addition to task execution and planning, project management involves managing uncertainty and adapting to change. This approach is particularly relevant in student projects, where learning, collaboration, and flexibility are central (Karlsen, 2021).

To guide the development process, the group adopted the Scrum framework, a widely used agile methodology that emphasizes iterative progress, continuous feedback, and adaptability (Scrum.org, n.d.). This framework made it easier to respond to stakeholder input, handle changing requirements, and keep the project moving forward in a focused way.

The overarching goal was to deliver a functional prototype within the constraints of the project. To manage this, the group worked within the framework of the project management triangle, a concept that highlights the relationship between three critical variables: scope, cost, and time. These three factors collectively influence the overall quality of a project (Coursera, 2025). The triangle illustrates that if one variable changes, the others must also be adjusted to maintain balance. For example, increasing the scope of a project typically requires either more time, more resources, or a compromise in quality.

In the context of this project, there were no direct monetary costs, so time effectively became the main limiting factor. As a result, the group had to carefully manage time and scope to maintain the desired quality within the available timeframe.

2.1 Project Management Methodology

The group followed an agile methodology based on Scrum, structured around one-week sprints, each with predefined goals and phases. This short sprint cycle supported frequent evaluation of progress and rapid adjustments to the project scope and technical implementation. For more details see section 8 Project Execution.

Furthermore, the following Scrum events were consistently followed (Scrum Guides, 2020):

- **Sprint Planning:** At the start of each sprint, the group defined sprint goals, broke tasks into manageable units, estimated effort, and assigned responsibilities based on availability and expertise.
- **Daily Scrum:** Short daily check-ins to ensure transparency and identify blockers early.
- **Sprint Review:** End-of-sprint presentations were used to showcase progress to the stakeholder.
- **Sprint Retrospective:** Internal evaluations were conducted to discuss what went well, what could be improved, and to agree on changes for the next sprint.

This structure promoted continuous improvement, team accountability, and an adaptable workflow. Additionally, the tasks were defined through brainstorming and analysis of project requirements. Larger tasks were broken into smaller items and added to a shared backlog in Azure DevOps, which also included assigned responsibilities and task priorities. While formal hour tracking was not used, this structure supported day-to-day organization and long-term planning. An overview of each sprint and the backlog can be found in Appendix 4 and 5 respectively. Additionally, a summary of the sprint retrospectives can be found in Appendix 13.

2.2 Risk Assessment

Risk management was an important part of the project planning process. At the start of the project, the group created a detailed risk matrix to identify and evaluate potential risks based on their likelihood and consequences. The matrix also included suggested mitigation strategies to prevent or minimize the impact of these risks. This approach aligns with Karlsen's (2021) emphasis on proactive risk identification as a key to maintaining project control and avoiding unexpected disruptions.

To assess risk levels, the group used a 1-4 scale for consequence and probability, generating a calculated risk score (Vicente, 2024). Each risk was given a consequence score (1 = minor, 4 = critical) and a probability score (1 = unlikely, 4 = very likely). Multiplying these values produced a risk score ranging from 1 to 16, which helped prioritize risks based on potential impact. These scores were then color-coded to indicate risk severity: green (1-4: low risk), yellow (5-9: moderate risk), and red (10-16: high risk). The full matrix, along with suggested mitigation strategies for each risk, is included in Appendix 6. A visual summary is provided in Figure 1 below.

Konsekvens	4	Moderat	Høy	Høy	Høy
	3	Lav	Moderat	Høy	Høy
	2	Lav	Lav	Moderat	Høy
	1	Lav	Lav	Lav	Moderat
		1	2	3	4
		Sannsynlighet			

Figure 1 - Example of Risk Matrix (University of Oslo, 2025)

Risk monitoring was also integrated into the team’s Scrum routines, particularly during daily scrum and sprint retrospectives. These regular check-ins allowed the group to identify emerging risks, reflect on ongoing challenges, and take corrective action when necessary. This continuous feedback loop was especially important in a time-limited and dynamic development environment, where agility and responsiveness were key to maintaining progress.

2.3 Quality Assurance

Quality assurance was an integral part of the project, embedded throughout the development process rather than treated as a final stage. The group aimed to ensure that the solution met both technical and user expectations, while maintaining alignment with stakeholder goals.

According to the International Organization for Standardization (ISO), quality assurance involves the processes and activities that provide confidence a product or service will fulfill quality requirements (iso.org, n.d.,). In project contexts, this means focusing on consistent practices that prevent errors and ensure that outputs meet defined standards. This view aligns with Karlsen’s (2021) perspective that project quality should be assessed based on “fitness for purpose,” meaning the product should fulfill stakeholder needs within the constraints of time, scope, and resources, rather than striving for unrealistic perfection.

This approach also reflects agile principles. The Scrum Guide emphasizes delivering usable, working increments frequently, valuing continuous improvements, and maintaining a shared

understanding of quality across the team (Scrum Guides, 2020). Quality is not viewed as a final checkpoint, but as something built into every step of the process.

In this project, quality was defined as the extent to which the product:

- Met the needs and expectations of stakeholder and end users.
- Provided a reliable, intuitive, and accessible user experience.
- Functioned consistently and without critical errors.
- Was well-documented for future development and maintenance.

To support and maintain an acceptable level of quality, the group implemented several strategies aligned with agile and industry best practices:

- **Code and design reviews:** Group members reviewed each other's contributions before merging code or integrating components, helping to catch errors early. This ensured that the system behaved as expected and met usability goals.
- **Manual testing:** The prototype was continuously tested, with particular attention to front-end elements and chatbot interactions. This ensured that the system behaved as expected and met usability goals.
- **Sprint retrospectives:** At the end of each sprint, the group reflected on what worked well and what needed improvement, both in terms of collaboration and technical output.
- **Stakeholder feedback:** Ongoing input from Kristiansand Municipality and the university supervisor ensured that development remained aligned with real-world needs.
- **Documentation:** Key architectural decisions, component functionality, and test results were recorded to ensure transparency and support future development in the KartAi project.

Throughout the project, the group also discussed what level of quality was realistically achievable given the time and scope constraints. Perfection was not the goal, but rather delivering a usable, functional and stakeholder-relevant product that fulfilled its intended purpose within the available timeframe.

2.4 Time Estimation

Although the group did not formally log hours, time estimation was an integral part of each sprint planning session. At the beginning of every one-week sprint, tasks were broken down, discussed, and roughly estimated based on complexity, priority, and group capacity. These estimates were made collaboratively, drawing on prior experience and familiarity with the relevant technologies.

Task planning and estimation were supported through Azure DevOps, which the group used to organize and visualize ongoing work (see section 7.1 Microsoft Azure DevOps). While no formal time-tracking tools or burndown charts were implemented, the backlog and task board served as a shared point of reference for workload distribution and sprint structure.

Responsibilities were assigned based on individual strengths and availability, which helped balance the workload and avoid bottlenecks. As the project progressed and new challenges emerged, time estimates were continuously adjusted, and tasks were reassigned when necessary to maintain momentum and ensure steady progress.

Since the project had no financial cost, time was effectively its primary resource. Efficient use of limited time was critical to achieving the desired level of quality. A summary of estimated time distribution by activity is included in Appendix 7.

3 Methods

This chapter outlines the methodological approach that guided the project from initial exploration through system analysis and design. The project was grounded in an agile, user-centered development process informed by stakeholder engagement, structured modeling techniques, and continuous feedback.

3.1 Overall Approach

The project followed an agile, iterative development model using the Scrum framework (Scrum Guides, 2020). Scrum was selected due to its flexibility and focus on incremental delivery, which aligned well with the project's exploratory and time-constrained nature. Additionally, the development process was guided by user-centered design principles, emphasizing usability, real-world needs, and active stakeholder involvement.

Object-oriented analysis principles as outlined by Mathiassen et al. (2018), provided the theoretical foundation for modeling the system. This approach emphasizes understanding the problem domain through stakeholder roles, system context, and functional responsibilities.

3.2 Data Collection and Stakeholder Involvement

The group collaborated closely with Kristiansand Municipality, the primary stakeholder. Data collection involved:

- **Stakeholder meetings:** Regular discussions with municipal representatives helped define the problem space and refine system requirements.
- **Observation sessions:** The group observed real guidance sessions between case workers and residents. These provided valuable insights into the challenges residents face in navigating the building application process.
- **Background research:** The group reviewed existing documentation from the KartAi project, as well as relevant regulations and digital services related to building applications from the Norwegian Directorate for Building Quality (DiBK).

These activities ensured that the analysis phase was informed by real user experiences and stakeholder priorities.

3.3 Modeling and Analysis Techniques

Several structured techniques were applied to synthesize findings and guide system design:

- FACTOR analysis (Mathiassen et al., 2018) was used to identify key elements of the system's context, roles, and responsibilities for both residents and case workers.
- User scenarios and information use cases were developed based on observations and stakeholder input to simulate real-world interactions and evaluate system usability (Sommerville, 2016).
- MoSCoW prioritization was employed to classify system features by importance and feasibility (Agile Business Consortium, n.d.). This allowed the group to manage the scope effectively and make informed trade-offs.

These methods supported an iterative analysis process, where new insights were continuously incorporated into evolving system requirements. A further detailed description of these methods will be given in Chapter 4.

4 Analysis

The purpose of the analysis phase was to define the system's functional goals, understand the context of use, and clarify stakeholder expectations. This phase laid the foundation for system design by identifying both technical constraints and user-centered requirements. The analysis was based on the methods outlined in Chapter 3, combining stakeholder insights, observations, and structured modeling

4.1 System Definition

The system definition for the project is as follows:

“The proposed solution is a real-time, web-based system designed to digitize and streamline the building application process. The system must allow residents to complete and submit building applications digitally, while supporting case workers in reviewing and processing them efficiently.”

The definition was shaped through stakeholder discussions and background research. The focus is to improve accessibility and usability for applicants, while simultaneously providing case workers with tools to reduce manual workload and ensure completeness.

Following Mathiassen et al. (2018), the group prioritized defining the system's responsibilities early to ensure alignment with user needs. The system serves as a bridge

between two user groups, residents and case workers, each with distinct roles, workflows, and expectations.

4.2 System Description

Based on the analytical framework from Mathiassen et al. (2018), the system is conceptualized as a dual-interface application:

- **Resident Interface:** A user-friendly, guided digital application flow. It provides access to regulatory information, form completion assistance, document upload, and map interaction tools.
- **Case Worker Interface:** An administrative view supporting AI-assisted case evaluation, such as identifying incomplete applications, verifying image and document quality.

While both interfaces are part of the system concept, the development of the project focused on the resident side due to resource constraints and identified user needs (see section 4.5 Scope Focus Based Analysis).

4.3 FACTOR Analysis

To understand the roles and responsibilities of each user group, a FACTOR analysis was conducted:

FACTOR	Case Workers Description	Residents Description
Functionality	Streamline case processing through automation and reduce administrative workload.	Provide a simplified and guided way to submit building applications digitally.
Application Domain	Case workers in municipal planning and building departments.	Residents or property owners applying for construction permits.
Context	Operates within a public administration environment requiring traceability, accuracy, and efficiency.	Users with limited experience in regulations or digital systems must complete legal applications independently.
Technology	Integration with AI tools in an existing or evolving web-based case system.	A responsive web application with integrated guidance, mapping, and document upload.
Objects	Digital building applications, municipal databases, internal guidelines.	Property data, regulations, form inputs, supporting documents, and uploaded images.
Responsibility	Ensure lawful, complete, and efficient application processing with support from automated tools.	Complete the application accurately, with the system providing proactive guidance and validation.

Table 2 - FACTOR analysis of Case Workers and Residents

4.4 User Scenarios and Prioritization

User scenarios were developed based on observed guidance sessions and stakeholder insights, following best practices from Sommerville (2016) and Mathiassen et al. (2018). These scenarios helped evaluate how the system could address real user challenges.

Use Case 1: Navigation and Information Retrieval

- **Scenario A:** A resident plans to build a garage and needs to understand legal requirements.
 - **Goal:** Determine if users can locate accurate regulatory information.
- **Scenario B:** A resident needs to find documentation requirements for building a terrace.
 - **Goal:** Evaluate how clearly documentation needs are presented.

Use Case 2: Application Creation

- **Scenario A:** A resident starts a new application to build a sunroom.
 - **Goal:** Assess the intuitiveness of the guided form.
- **Scenario B:** A resident adds a map section to mark building location.
 - **Goal:** Test the precision and usability of the map interface.

4.4.1 MoSCoW Prioritization

To manage the scope, the group applied the MoSCoW prioritization method (Agile Business Consortium, n.d.):

- **Must have:** Critical features without which the system is unusable.
- **Should have:** Important features that are not essential but add significant value.
- **Could have:** Desirable features that enhance user experience but are not time-critical.
- **Won't have:** Features intentionally excluded from the current scope due to time, complexity, or resource constraints.

This framework allowed the group to prioritize critical features while maintaining flexibility and realism. Based on this method, the following priorities were defined:

- **Must have:** Guided form flow, document upload, map interaction, and basic chatbot support.
- **Should have:** Real-time validation of missing data, progress indicators, and confirmation prompts.
- **Could have:** Save draft functionality, and multilingual interface.
- **Won't have:** Full integration with national databases or automated decision-making features.

These priorities ensured that the group stayed focused on delivering core functionality within the available timeframe and capacity.

4.5 Scope Focus Based Analysis

Following the analysis phase, the group made a strategic decision to focus primarily on the resident-facing side of the system. While both user groups, residents and case workers, were considered, several factors influenced this prioritization.

First, the analysis indicated that residents faced most significant challenges in understanding and completing the building application process. This was especially true in relation to legal requirements, documentation standards, and general digital literacy. Observations from guidance sessions reinforced these findings and emphasized the need for more accessible and supportive digital solutions for residents.

Second, with only three group members and a limited timeframe, the system had to be scoped realistically to ensure the delivery of a functional and testable prototype. As noted by Mathiassen et al. (2018), the project scope should be refined iteratively in relation to available resources and stakeholder readiness.

Additionally, the municipal case workers had recently begun using a new internal system and had limited experience with its workflows. Further, the group was informed that it typically takes up to two years for new case workers to become fully comfortable in their roles. Thus, designing features for an environment still in flux would therefore have been premature and potentially misaligned with real needs.

These combined factors supported the decision to prioritize the resident interface. The final prototype centers on a digitized application flow for residents, incorporating selected KartAi technologies such as image validation and chatbot support to improve application quality and reduce the workload for municipal case workers downstream.

5 Design

This chapter outlines the design process behind the prototype, developed with a user-centered approach to improve the experience of residents submitting building applications. Following the analysis phase, the group focused on usability, clarity and guidance, using established design frameworks and principles.

User experience (UX) design involves shaping not only the interface and interaction but also the emotional, contextual, and long-term aspects of user engagement (Benyon, 2019). This holistic view influenced the group's decisions across layout, functionality, and feedback mechanisms. This section details the user interface design, key usability strategies, supporting interaction frameworks such as PACT and Benyon's design principles, as well as the system's navigation structure, wireframes, and core accessibility considerations.

5.1 User Interface

The user interface (UI) was designed to support residents with varying degrees of digital literacy and no assumed knowledge of building regulations nor the building application process. It guides users through a linear application process that includes selecting application type, uploading required documents, and placing the project on a map. The group has designed supporting pages outside the core application flow, including guidance and information pages, a checklist, and other resources. These elements provide users with contextual help and preparatory information to improve understanding and reduce friction during the application process. The interface emphasizes:

- Minimalist design with focused tasks per screen.
- Progressive disclosure of information to reduce cognitive overload.
- Clear buttons and icons that afford interaction.
- Immediate visual feedback on inputs and errors.

This reflects Benyon's (2019) view of user experience as an ongoing journey where consistency, clarity, and feedback contribute to a smooth and confident interaction.

5.2 PACT Analysis

The PACT framework; Peoples, Activities, Context, Technologies, introduced by Benyon (2019), was applied to analyze the interactive space. This helped identify realistic user expectations and environmental constraints.

PACT Element	Design Considerations
People	Residents with diverse digital and legal understanding, meaning that inclusive design was required.
Activities	Completing forms, uploading files, using a map, reading guidance, meaning that it must be split into clear and sequential steps.
Context	Use from home on various devices, often without technical support, which means that reliability and clarity are essential.
Technologies	A responsive web-based system compatible with major browsers and accessible on PCs.

Table 3 - PACT Analysis

The analysis confirmed the importance of responsiveness, step-by-step support, and system forgiveness (undo, edit, confirmation prompts).

5.3 Benyon's 12 Design Principles

To guide the interface development, the group evaluated and prioritized Benyon's 12 design principles (Benyon, 2019). These principles offer a comprehensive framework for creating interfaces that are learnable, effective, and accommodating. Each principle was assessed in light of the system's intended users, the complexity of the building application domain, and the project's time and technical limitations. The group discussed how each principle aligned with observed user needs and stakeholder feedback and categorized them accordingly. The resulting prioritization is shown in Table 4.

Benyon's 12 Design Principles					
Category	Principle	Very Important	Important	Less Important	Irrelevant
Learnability	Visibility	X			
Learnability	Consistency		X		
Learnability	Familiarity	X			
Learnability	Affordance	X			
Effectiveness	Navigation	X			
Effectiveness	Control		X		
Effectiveness	Feedback	X			
Effectiveness	Recovery	X			
Effectiveness	Constraints	X			
Accommodation	Flexibility			X	
Accommodation	Style			X	
Accommodation	Conviviality	X			

Table 4 - Benyon's Design Principles Prioritization

5.3.1 Design Implications

The very important principles were prioritized due to the complexity of the building application process and the diverse user base. For example:

- Visibility and Affordance ensured that buttons and features clearly suggested their use and reducing user uncertainty.
- Familiarity was reinforced through consistent layout and labeling patterns based on common public service interfaces.
- Navigation, Feedback, and Recovery were essential in preventing user frustration and reducing application errors.
- Constraints (e.g., required fields, input validation, etc.) were implemented to help users avoid mistakes before submissions.
- Conviviality supported a tone of communication that was friendly and helpful, without being overly technical.

The important principles, Consistency and Control, supported learnability and gave users confidence that the system would behave predictably. Finally, Style and Flexibility were considered less important due to time constraints and the public-sector context, where simplicity and clarity were prioritized over personalization or aesthetic customization. This prioritization helped the group stay focused on what mattered most for the user experience, especially with limited development resources.

5.4 Navigation Map

The navigation structure of the prototype is designed to provide users with a consistent and intuitive way to access key features of the system. A persistent sidebar menu is used for primary navigation, offering clear pathways to each step of the application process. This ensures a shallow navigation depth and aligns with Benyon's design principle of supporting efficient and predictable navigation.

The sidebar contains the following elements:

- **Min Oversikt** (*My Overview*): A personal dashboard summarizing active applications and their statuses, along with useful shortcuts and reminders.
- **Før du søker** (*Before You Apply*): An introductory section with guidance, requirements, and regulations that apply to different types of applications.
- **Sjekkliste** (*Checklist*): A dynamic list showing which tasks are complete, what documents are required, and what remains during the different phases of submission.
- **Min Eiendom** (*My Property*): General information about the user's property.
- **Mine Søknader** (*My Application*): A list of previously submitted and saved applications with status updates and timestamps.
- **Mine Dokumenter** (*My Documents*): A document management section for uploading and reviewing all attachments.
- **Meldinger** (*Messages*): A communication area where users receive updates, feedback, and requests from case workers and the system.

In addition to the sidebar, there are some other features in the navigation bar, for example a log-out button is always available in the top right corner of the interface, allowing users to safely exit the session. This promotes security and follows user expectations for account-based systems.

This navigation layout supports flexibility while maintaining a linear, guided experience. Users can jump between sections when needed without losing orientation, and each page clearly displays the current step. A visual navigation map is included in Appendix 8 to illustrate the system structure and flow.

5.5 Wireframes

Using Figma, the group created low- to mid-fidelity wireframes to define layout and test early design decisions. The wireframes included:

- Structured forms with clear labels.
- Document upload zones with visual cues and drag-and-drop capability.
- A simplified integration of AI technologies.

The wireframes were evaluated internally and shared with the stakeholder for feedback, and the insights gathered shaped both visual style and user flow.

5.6 UX: Accessibility, Usability, and Acceptability

Benyon emphasizes that good UX goes beyond usability, it must also be accessible and acceptable (Benyon, 2019). The system was evaluated through these lenses:

- **Accessibility:** Design choices considered Web Content Accessibility Guidelines (WCAG) which are directions for how web pages should be designed regarding adequate contrasts, font sizes, and keyboard navigation. Elements were tested on screen readers to ensure compatibility.
- **Usability:** Input fields, forms, and buttons followed conventional design patterns. Tasks were short and clearly segmented. Feedback helped users understand errors and how to fix them.
- **Acceptability:** The system was made visually clean, branded appropriately to the public sector, and designed to build user trust through transparency and a structured process. Acceptability was further supported by avoiding unnecessary technical jargon and ensuring users felt in control throughout the application.

These principles were particularly important given the diversity of users in public services and the high cognitive demands of building applications.

6 Architectural Design

This chapter outlines the architectural design of the system and highlights the core priorities and decisions that shaped it. A solid architecture provides the foundational structure of any software project, supporting clarity, scalability, and maintainability. As emphasized by Mathiassen et al. (2018), an explicit architectural overview helps groups understand the system structure, coordinate work across components, and ensure quality throughout development.

In this project, the architecture was defined early to ensure that the technical structure aligned with the system's functional goals. The architectural design enabled the group to divide tasks effectively and ensure flexibility for future enhancements. This chapter begins by presenting

the design criteria in Table 5 and continues with an overview of the processes and components that make up the system.

Table 5 presents a prioritized overview of 12 design criteria relevant to software architecture, based on the framework by Mathiassen et al. (2018). The group ranked each criterion based on stakeholder needs, system goals, and internal discussions. In some cases, the criteria were inherently fulfilled due to standard development practices.

Criteria	Very important	Important	Less important	Irrelevant	Easily fulfilled
Usable					X
Secure	X				
Efficient					X
Correct		X			
Reliable	X				
Maintainable	X				
Testable					X
Flexible		X			
Comprehensible	X				
Reusable	X				
Portable			X		
Interoperable	X				

Table 5 - Prioritized Architectural Design Criteria

6.1.1 Very Important

- Secure:** The group assumed that security would be a key concern, especially considering potential future use in a real municipal setting. After deliberating with the stakeholder, it was decided that the implementation of security was outside of the project scope, due to the group's capacity, and security would be thoroughly implemented in future iterations. Although the system does not currently process personal data, architectural decisions were made to ensure security could scale, including secure hosting (Azure VM), separation of front-end and back-end layers, and version-controlled deployment.

- **Reliable:** The system had to work consistently and without failure. This was ensured through stable component selection (e.g., React, Next.js), code modularity, and regular manual testing.
- **Maintainable:** With only three developers and potential handoff after the project, maintainability was prioritized. The architecture is modular, using clear folder structures, and reusable components to allow future developers to easily understand and extend the system.
- **Comprehensible:** Closely tied to maintainability, comprehensibility was emphasized by documenting API structure, component usage, and deployment flow. This increases transparency and simplifies knowledge transfer.
- **Reusable:** Components were built with reusability in mind. For instance, form elements and UI modules were designed generically to be reused across multiple screens or adapted to other use cases within KartAi.
- **Interoperable:** The system was developed to integrate KartAi's AI technologies, such as the chatbot and document analysis features. This required careful planning of data formats and interface points to ensure clean and scalable integration with external services.

6.1.2 Important

- **Correct:** The system must handle data properly and guide users toward correct application submission. While not the top-ranked priority, correctness was supported through validation functions, controlled form flows, and error detection (e.g., required fields and accepted file types).
- **Flexible:** Flexibility was important to allow future extensions of the system, such as integrating more AI features or supporting more application types. The system's modular design supports this, though the limited project scope meant some flexibility-related features were not fully implemented.

6.1.3 Less Important

- **Portable:** Since the product is a Proof of Concept, portability was not a primary concern. However, this requirement had already been fulfilled, as the system the group inherited included Docker support, making it portable by default. Additionally, the final product is currently being hosted on Azure.

6.1.4 Easily Fulfilled

Some criteria were considered essential but did not require special attention due to standard development practices:

- **Usable:** Usability was a non-negotiable part of the project's scope and was addressed extensively in the design and testing phases (see 5 Design and 9 Testing and Evaluation).
- **Efficient:** While not optimized for performance, the use of lightweight technologies and modern frameworks (React, Next.js) ensured baseline efficiency without additional architectural decisions.
- **Testable:** Manual testing was performed throughout development. Due to the clean separation of logic in components and use of simple APIs, the system was inherently testable. Automated testing was not prioritized due to time constraints.

6.2 Processes and Components

The system is designed to assist users in filling out building applications. This is achieved through a chatbot, powered by a Large Language Model (LLM), that guides the user by providing relevant information about the regulations and municipal plans. The chatbot operates as an agentic system and has access to three main tools: (1) Document Search - retrieves information from a vector database containing general building regulations, municipal plans, and internal notes from guidance meetings between applicants and case workers. (2) Internet Search - searches online sources for additional relevant information not covered in the internal database. (3) Spatial Analysis - analyses geographic and zoning data to determine whether construction is permitted in the user's desired location.

The chatbot is integrated with the TiltaksAiD API, a KartAi tool which provides a map interface where users can mark the placement and size of their planned building directly on a map of their property. This spatial information is used in the analysis to identify regulatory requirements or potential conflicts.

Figure 2 illustrates the architecture of the chatbot and the flow of information. First, the user sends a prompt to the agent. The agent, powered by a LLM, then decides whether it needs to call one of the available tools, mentioned above. If a tool is needed, the agent calls the corresponding API and retrieves the result. Afterwards, the agent evaluates whether it has enough information to reason and generate a response, or whether it needs to call another tool to gather more information (Aradhyula, 2025).

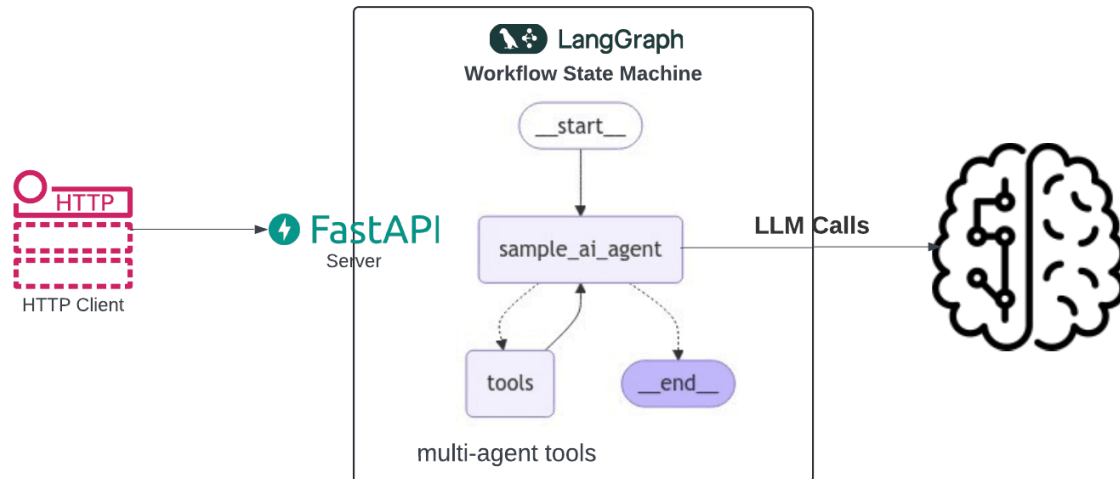


Figure 2 - Chatbot Architecture (Aradhyula, 2025)

The rest of the application provides a clear workflow for completing a building application. This is achieved through a React-based front-end, which interacts with the back-end to save and manage the user's input data.

The front-end is built as a form-based interface, where users can manually fill out their building application. All data is sent through the API endpoints defined in TypeScript Remote Procedure Call (tRPC), where it is processed and stored. The system allows users to close the application without completing it, and later return to edit, review, and finalize the application before submission.

Figure 3 represents the system architecture of the final product.

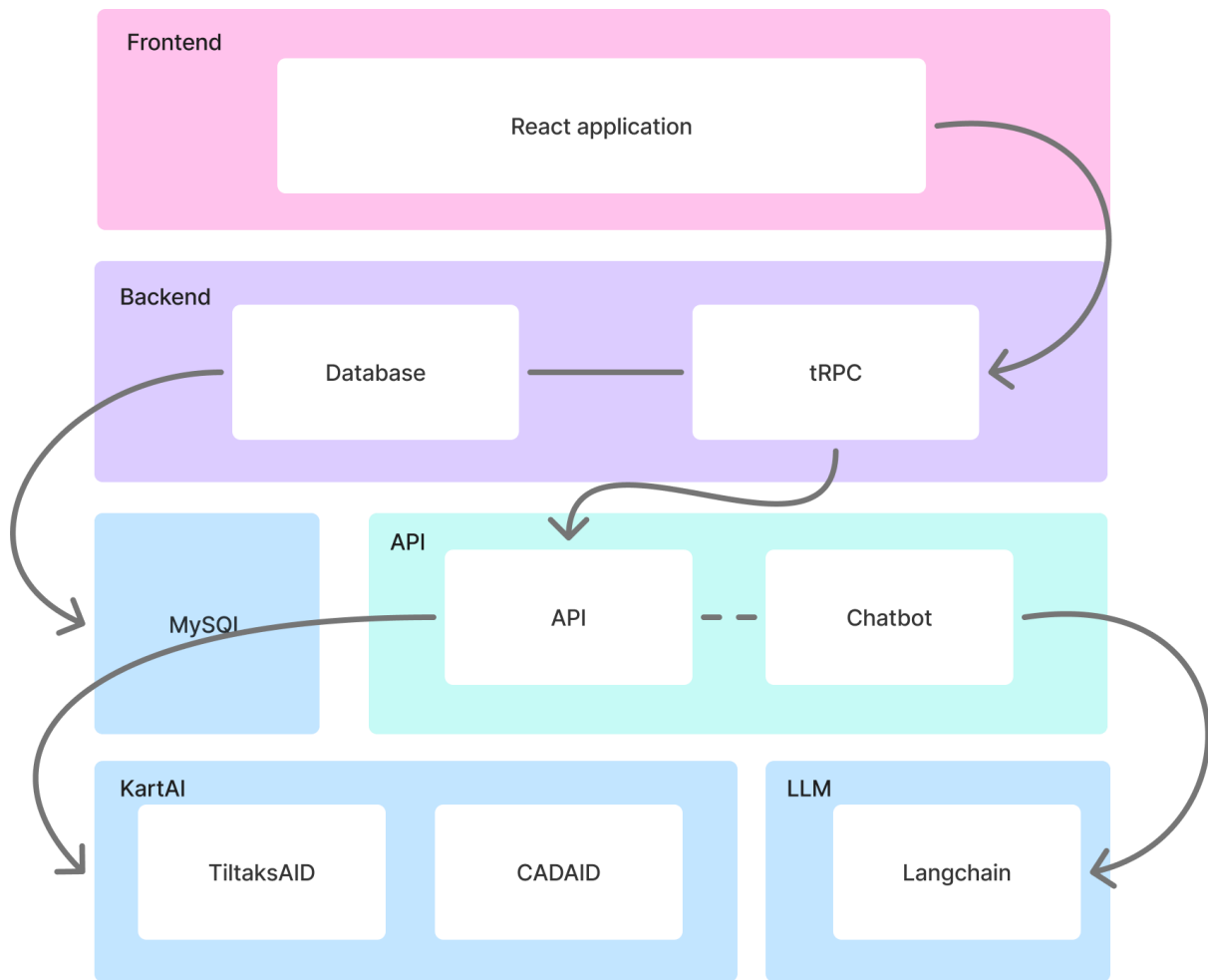


Figure 3 - System Architecture and Information Flow

7 Technology and Tools

This chapter outlines the technologies and tools used during the development of this bachelor project. Tool selection was driven by a need for efficiency, collaboration, and compatibility with existing systems. The tools covered here supported everything from planning and design to implementation and deployment, allowing the group to work effectively in a small group and within the project's time constraints.

7.1 Microsoft Azure DevOps

Azure DevOps was selected as the group's project management tool. Although its full range of features was not fully utilized, it served as a platform for maintaining a product backlog and organizing development tasks. The group primarily used the Boards feature, a digital kanban board system that allows teams to create, assign, and track work items. Boards support agile workflows by visualizing tasks across columns such as "To Do," "In Progress," and "Done," making team progress and responsibilities easy to monitor.

Azure DevOps also helped align the group with the Scrum methodology, providing visibility into sprint goals, task progress, and pending items. While no exact time estimates or burndown charts were implemented, the backlog served as a shared reference for workload distribution and sprint structure.

7.2 Figma

Figma was used extensively for the design and prototyping phase of the project. As a collaborative design tool, it allowed the group to create wireframes, mockups, and interface components that reflected the layout and flow of the intended system. Figma's cloud-based interface enabled real-time collaboration and feedback, both internally and with stakeholder.

The tool was especially useful for iterating on design ideas, visualizing user journeys, and aligning UI structure with the principles outlined in the design chapter (see 5 Design). It also served as a bridge between the design and development phases, as developers could inspect styles and layouts directly. The Figma workspace can be found on Appendix 9.

7.3 Programming Languages

This section aims to explain the programming languages which have been used in this project. Since this project is a continuation of the established Proof of Concept (PoC), the foundations of what technologies to use were already laid out for the group. Because of this, the group decided to continue using the current technologies that have already been implemented, as converting the whole project into a different programming language would take too much time and overall be unnecessary.

7.3.1 T3 Stack

T3 Stack contains and combines a handful of useful technologies for web app creation, such as React, Next.js, TypeScript, TailwindCSS, and tRPC. This framework has effectively created a reliable and dynamic way of further developing the full-stack PoC. The NTNU students mention in their report that the reason for choosing this framework was due to the familiar technologies it contains, allowing for a quick start on the developing, and reducing their learning curve (Aarø et al., 2024). We were not entirely familiar with these technologies but decided to continue with them to avoid the time-consuming effort it would take to rewrite the project in a different language.

7.3.2 React

React is a very popular library written in JavaScript, made for creating and building user interfaces. One of the main features of React, is that it allows for the creation of “reactive”

components. These components can then be updated based on a user's interactions and behavior.

7.3.3 Next.js

Next.js is a React-based framework efficient for building full-stack web applications. Among its features are Server-Side Rendering (SSR), Static Site Generation (SSG), dynamic routing, and automatic code splitting (NEXT.js, n.d.). These features, among others, make web applications easier to scale and maintain.

7.3.4 TypeScript

Programming in TypeScript allows for maintainability and improves code reliability. It also enables a reduction in runtime errors with static type-checking, which also facilitates smoother collaboration with other developers by making the code itself more self-documenting.

7.3.5 TailwindCSS

TailwindCSS is highly flexible, allowing for the maintaining of consistent, clean and responsive design. Additionally, it can instantly deploy new layouts or changes to the style, mainly due to the abundance of built-in methods and pre-defined breakpoints.

7.3.6 tRPC

tRPC is a framework which allows for seamless communication between the front-end and back-end parts of the project, by building type-safe APIs which is directly integrated into the T3 Stack. A feature of tRPC, is that it can help avoid redundancy by defining and invoking back-end processes without creating additional API routes.

7.3.7 Prisma Object-Oriented Mapping

Prisma ORM allows for a more intuitive and code-driven way of defining a database schema. This makes the database easier to manage and interact with the tables. Prisma auto-generates TypeScript types from the schema, ensuring type safety across the application.

7.3.8 MySQL

MySQL is a database which is very compatible with the technologies being implemented. It is highly scalable and provides a reliable structure for organizing and managing data, fitting the application's need for ample storage.

7.3.9 Python

Python is a very popular language to use when developing any tools relating to AI. Additionally, an AI chatbot developed with Python will make the integration into the other KartAi agents more seamless.

7.4 GitHub

GitHub was used for version control and team collaboration. It provided a reliable and structured way to manage source code, track changes, and collaborate across features. The group worked with branches to separate development efforts and avoid merge conflicts, although advanced workflows (e.g., CI/CD pipelines or automated testing) were not implemented. GitHub was also used to maintain project transparency, support backup routines, and ensure that all changes were documented and reversible.

7.5 APIs

Application Programming Interfaces (APIs) are a central component of the project and can be defined as “how the features and services in one software system can be accessed by another” (Chevre, 2024). The group's main objective was to integrate the existing KartAi APIs into a building application to simplify the process for users. APIs allow different software systems to communicate with each other by defining how their features and services can be accessed.

From the many available KartAi APIs, only TiltaksAiD and CADAiD were selected for this project. This was a deliberate decision as the group with the approval of stakeholder decided that including too many AI technologies could confuse users. This would be the opposite of the project's goal, to create a more intuitive and user-friendly application.

The selection of APIs for the system was directly informed by a data analysis conducted by DiBK, which investigated the main reasons why case management for building applications tends to be so time-consuming and resource intensive. According to the analysis, two of the biggest challenges are missing documentation, and the lack of understanding of the requirements (Steinnes, 2023).

Dataanalyse av mangler

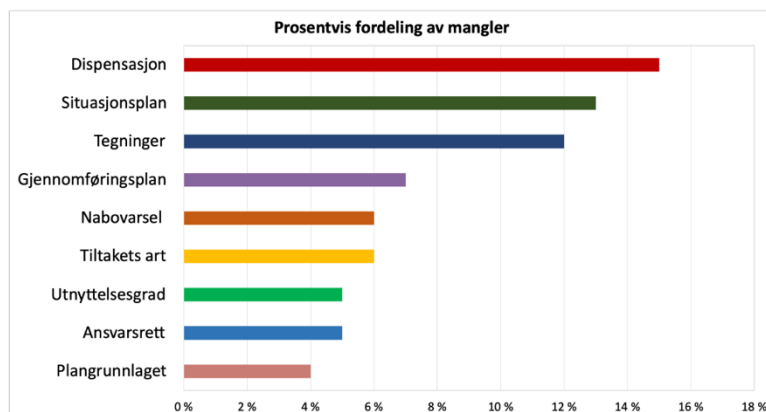


Figure 4 - Data Analysis from DiBK (Steinnes, 2023)

Figure 4 shows a breakdown of the most common deficiencies in submitted applications. The top three issues are: (1) Dispensation, (2) Site Plan and (3) Drawings. Together, these categories alone account for approximately 40% of all application deficiencies.

This analysis highlights that a significant portion of the delays and rejections stem not from technical issues, but from incomplete or incorrectly filled documentation, which can be mitigated by better guidance and digital tools. For instance:

- CADAiD helps users understand whether their drawings and sketches meet the formal requirements, by giving immediate feedback on what is missing or incorrect. This directly addresses the common issue of missing situation plans and drawings, which often result from users not knowing what to submit or how to produce it.
- TiltaksAiD offers real-time measurements and visual feedback on zoning rules, showing exactly where construction is permitted. This helps users avoid misunderstandings related to dispensation requirements and ensures they check against planning regulations before submitting the application.
- Although still under development, TiltaksAiD also shows potential for reducing omissions related to implementation plans, neighbor notifications, and responsibility declarations, by offering step-by-step guidance during the form-filling process.

By integrating these two APIs, the application helps users overcome major barriers in the building application process, such as understanding regulations and ensuring complete documentation.

For the chatbot implementation, two additional APIs were required: one for the Large Language Model (LLM) and one for embedding generation. The LLM API is used to ensure that the chatbot's text generation is natural and human-like, rather than robotic. This approach is important because the goal is for the chatbot to potentially replace or complement a

building application advisor, making it easier for users to understand the regulations and application process. The Embedding API was integrated to improve the accuracy of the chatbot's information retrieval. By converting documents into vector representations, the system can retrieve the most relevant information for a given user query, ensuring that the chatbot provides precise and helpful answers.

7.6 Microsoft Azure VM

The application was deployed to a Microsoft Azure Virtual Machine using Docker containerization. This allowed us to provide an environment for testing by municipal workers. The main goal of this deployment was to gather valuable feedback for further improvements of the application.

Although the deployment occurred relatively late in the development process, it is still highly beneficial. It enabled real-world testing outside the development environment and provided insights that are useful both for final adjustments and for future developers who may continue the project.

8 Project Execution

This chapter outlines the practical execution of the bachelor project across 17 one-week sprints. Building on the agile Scrum methodology, each sprint involved planning, development goals, and retrospectives. The workflow was iterative and collaborative, with a continuous focus on user needs and stakeholder feedback. The project was divided into five key phases, each with specific objectives and deliverables.

8.1 Beginning of the Project

The project began informally in mid-December 2024 with an introductory meeting between the stakeholder, Kristiansand Municipality, and the student group. During this session, the stakeholder presented the broader KartAi initiative along with an existing PoC developed by students from NTNU, which the group was tasked with further developing. Initial discussions made it clear that the stakeholder aimed to move beyond a conceptual showcase of KartAi technologies and toward a functional prototype that demonstrated their practical application.

When the project officially commenced in January 2025, the group initially found the scope to be undefined. This was a deliberate choice by the stakeholder, who wished to grant the group flexibility in determining the project's direction, provided it remained focused on implementing KartAi technologies in a meaningful, real-world context. After conducting various analyses and engaging in discussions with the stakeholder, the group chose to pursue an approach centered around the citizens.

During the KartAi kick-off meeting held at the end of January, the stakeholder and the university supervisor provided valuable guidance that helped the group formulate a clear problem statement and a set of research questions for the project.

8.2 The Sprints

The project was divided into 17 one-week sprints, structured into five distinct development phases, followed by a final post-sprint period for continued work and final touches before the exam. Each phase was aligned with specific goals and outcomes that guided the progression from early analysis to a completed, functioning prototype. The phases were as follows:

1. **Exploration Phase:** Understanding the task and defining the problem space.
2. **Design and Concept Phase:** Ideating, wireframing, and establishing the user interface.
3. **Core Development Phase:** Building the foundation and integrating key functionality.
4. **Refinement Phase:** Extending functionality and improving usability.
5. **Finalization Phase:** Completing core development and preparing for delivery.
6. **Continued Work Phase:** Deployment, documentation, and usability testing before exam

Each of the following sections describes these phases in more detail and summarizes what was achieved during the corresponding sprints.

8.2.1 Pre-Sprint – Sprint 1 (02.01.25 – 24.01.25)

Phase: Exploration

The first phase of the project focused on understanding the problem domain and clarifying the project's direction. During this period, the group engaged in discussions with the stakeholder, reviewed the existing Proof of Concept developed by the NTNU students, and familiarized themselves with KartAi technologies. This stage also involved conducting analysis activities, such as stakeholder mapping, FACTOR and PACT analyses, and identifying potential user needs. The group defined initial requirements and created a shared backlog using Azure DevOps. This groundwork allowed the group to enter the design phase with a clearer understanding of the problem they were solving.

8.2.2 Sprint 2 – Sprint 5 (25.01.25 – 21.02.25)

Phase: Design and Concept

In this phase, the group transitioned from research to ideation and design. Several interface proposals were developed using Figma, informed by earlier analysis and stakeholder input. These design concepts aimed to improve user accessibility and streamline the building application process. By the end of the phase, the group had selected a preferred design

direction and initiated early development. Basic front-end structures were implemented using React, including foundational components and the initial navigation layout. This phase helped bridge the gap between conceptual design and practical implementation.

8.2.3 Sprint 6 – Sprint 10 (22.02.25 – 28.03.25)

Phase: Core Development

During this core development phase, the group focused on building key features of the prototype and integrating system components. The front-end and back-end were connected using Node.js and Prisma ORM, enabling the processing of form inputs, file uploads, and document validation. The group also continued refining their design in Figma to reflect the evolving structure of the working system. A significant milestone during this phase was the idea pitch to stakeholder (see Appendix 10), where the group proposed integrating a chatbot to support users throughout the application process. This idea was developed through all the background research and participation in the observation sessions (see Appendix 11), where the group noticed a need for better guidance throughout the building application process. Thus, the idea of 24/7 guidance which knows the laws and regulations and a mapping tool for residents to see if they must or/and to help them through the difficult process. By the end of this phase, the system's main application flow was functional and ready for refinement.

8.2.4 Sprint 11 – Sprint 14 (29.03.25 – 25.03.25)

Phase: Refinement

The refinement phase involved expanding functionality and improving the overall user experience. The chatbot was integrated into the system and tested for its ability to assist with regulations and documentation queries. At the same time, the group improved form validation, error messaging, and UI responsiveness based on internal testing and feedback. Additional features such as the checklist and message center were added to simulate a more complete building application workflow. Technical documentation and architectural diagrams were also updated during this phase to prepare for the final delivery.

8.2.5 Sprint 15 – Sprint 17 (26.04.25 – 16.05.25)

Phase: Finalization

In the final three sprints of the formal development cycle, the group focused on polishing the product and preparing for delivery. Remaining bugs were fixed, styling was adjusted for clarity and accessibility, and the prototype was prepared for deployment on Microsoft Azure Virtual Machine for stakeholder testing.

During this period, the group also finalized the written report and supporting documentation for submission on 16.06.25, the official deadline for the bachelor project. This included

detailing the system design, analysis, development process, and key reflections. Due to unresolved challenges with Azure deployment, the group planned to continue addressing technical issues and conduct informal user testing after submission, as part of their ongoing commitment to the KartAi project.

By the end of Sprint 17, the prototype showcased AI-enhanced functionality within a guided, user-friendly interface, aligned with the original goal of transforming the PoC into a more applied, practical solution.

8.2.6 Sprint 18 – Exam Period (17.05.25 – 02.06.25)

Phase: Continued Work and Final Touches

Although the structured sprint cycle concluded with Sprint 17, as of the official submission of this thesis, the group will continue to make adjustments and refinements to the project during the period leading to the final exam. This final phase focuses on completing outstanding tasks, including finalizing documentation, resolving deployment challenges, and performing limited user testing.

A key focus has been creating a stable development environment on Microsoft Azure Virtual Machine (VM). The group encountered ongoing technical issues since Sprint 15 and will use this period to continue troubleshooting and testing to ensure the system can be accessed and demonstrated reliably.

In parallel, work on the final version of documentation will continue. The group also planned informal usability testing with relevant users, with the aim of gathering last-minute feedback on functionality, clarity, and user flow. The tasks designed for user testing are included in Appendix 12. If feasible, insights from this testing may be presented during the exam.

The final period reflects the group's continued commitment to delivering a complete, user-centered prototype supported by well-documented work, in line with the academic expectations and the goals of the KartAi project.

8.3 Collaboration and Meetings

Throughout the project, the group maintained a structured and consistent collaboration routine with both internal group members and external stakeholder. This approach combined regular agile practices with formal checkpoints to support steady progress and alignment with the project's goals.

Weekly update meetings were held with the stakeholder every Friday. These informal meetings created a continuous feedback loop, allowing the group to share progress, clarify questions, and adapt development priorities. In addition, the group worked from the municipality's office every Friday, with extra in-person workdays added as needed. This regular physical presence helped strengthen communication, build rapport with the stakeholder, and create a more integrated and collaborative working environment.

Internally, the group followed Scrum principles by conducting sprint retrospectives at the end of each of the 17 one-week sprints. These meetings provided space for reflection, process improvement, and task reassignment based on progress and challenges. Common topics included collaboration effectiveness, workload balance, technical roadblocks, and sprint outcomes. A summary of recurring themes and takeaways from the retrospectives is included in Appendix 13.

To ensure structured feedback and maintain alignment with the stakeholder's expectations, the group held three formal steering group meetings (styringsgruppemøter) with the representatives from Kristiansand Municipality. These meetings were conducted on 07.02, 04.04, and 12.05, aligning with key project milestones. Each session included a demonstration of the current prototype, discussion of challenges and opportunities, and planning for the next development phase. Full meeting minutes (møtereferat) are included in Appendix 14.

In addition to the internal stakeholder meetings, KartAi demo meetings (demomøter) were held approximately every other Friday as part of the broader KartAi project. These sessions involved multiple partners and project contributors. The group occasionally presented updates on their work during these meetings, particularly in the early and middle stages of the project. However, due to time and capacity constraints during later sprints, participation in the demo sessions became less frequent as the group focused on completing core development tasks.

The group also participated in two obligatory status presentations (Status 1 and Status 2) organized by the University of Agder as part of the IS-304 course. These formal presentations provided academic checkpoints to demonstrate progress, receive feedback, and reflect on development decisions. Both were attended by the course coordinators and fellow student groups. Presentation materials are included in Appendix 15.

This combination of informal collaboration, structured internal reflection, stakeholder coordination, and academic checkpoints contributed to a responsive and well-aligned development process.

9 Testing and Evaluation

This chapter describes how the developed prototype was tested and evaluated throughout the project. Due to the limited timeframe and resources, the group relied primarily on manual testing methods and automated evaluation tools to ensure usability, performance, and basic accessibility. While large-scale user testing was not feasible, iterative testing and stakeholder feedback played a crucial role in identifying and addressing usability issues and functional bugs. The focus on testing was to verify core functionality, validate chatbot responses, and ensure the solution aligned with usability and accessibility expectations for a public-facing application.

9.1 Google Lighthouse

To evaluate the technical quality of the front-end application, the group regularly used Google Lighthouse, an open-source tool integrated into Chrome DevTools which highlights where improvements can be made. Lighthouse provides automated audits across several categories, including Performance, Accessibility, Best Practices, and Search Engine Optimization (SEO). Additionally, a score out of 100 is provided in each of the categories to summarize the audits. For this project, the Accessibility and Performance scores were emphasized, as the application was designed for a diverse set of users, including those unfamiliar with digital platforms.

The results of the Lighthouse test (full report in Appendix 16) reported an Accessibility and SEO score of 100, a Performance score of 94, and a Best Practices score of 78. For Performance, areas of improvement were identified to be large layout shifts which can be disorientating for users, and the prevention of back/forward cache restoration on the pages. In the Best Practices category, the biggest issues were the use of third-party cookies, and minor attributes associated with certain elements in the code.

Since Lighthouse operates on automated checks, it is not unreasonable to think that the tool might miss or overlook some technical nuances which require manual testing.

9.2 Manual Testing of Chatbot

Testing of the chatbot was an ongoing activity aimed at ensuring that the guidance it provided was both helpful and accurate. As the project followed an agile development methodology, testing was performed continuously throughout the development phase rather than only at the end. To support better tracking and analysis, the LangSmith platform provided by LangChain, was used to gain detailed insights into each step the agent took, including which tools were used, the internal thought process, and any runtime errors encountered (LangChain, n.d.,-a). Additionally, it gave statistics on token usage and timing for each action, making it easier to identify bottlenecks and track down the causes of failures.

The primary method for manual testing was writing and submitting example prompts that a typical user might send. This allowed the team to observe how the agent interpreted different types of input, whether it chose the correct tool, and how it reasoned its way to a final answer. The LangChain framework's ReAct Agent process, which stands for "Reason" and "Action", was a valuable resource, as it provided detailed logs of the agent's decisions and actions. These logs were crucial for diagnosing issues, especially runtime errors and incorrect behavior (LangChain, n.d.,-b).

Throughout the testing period, several challenges were discovered that highlighted the importance of continuous testing. Among the main issues were hallucinations, difficulties in guiding the abstract reasoning of the ReAct Agent, incorrect tool usage, and unnecessary iterations leading to higher token usage and runtime errors.

To address these problems, several key improvements were made. A fallback response was introduced to handle failures more gracefully. The number of allowed reasoning steps was limited to prevent excessive token consumption and reduce errors. In addition, the database content was improved to provide the chatbot with more accurate and relevant information, which helped reduce hallucinations and improve the overall reliability of the guidance.

Overall, as this was the group's first time developing an agentic chatbot, continuous testing proved essential. It helped uncover unexpected challenges and guided key improvements in both functionality and user experience.

10 Final Product

The final product is a prototype of a digital solution designed to assist users throughout the building application process. The system includes a chatbot that takes on the role of an advisor, as well as a form-based interface for submitting building applications. Although the product is not yet complete, it showcases the core functionality such a system should include. This is made possible through the integration of two existing KartAi technologies: CADAiD and TiltaksAiD.

This chapter will further present the current functionality of the application, the foundation for further development, and future plans.

10.1 Functionality

The functionality of the application is defined by what can currently be tested by the users. While in theory all components of the system can be tested, the focus has been on the core elements that directly support applicants throughout the building application process.

The overall goal for the system is to support the three main stages of a building application: before, during, and after submission. The current solution fully covers the before and during phases, with partial implementation of the after phase.

The before phase aims to prepare the user by providing the necessary information and guidance for submitting a building application. To support this, the system includes an overview page, a checklist, and access to key resources that explain regulations, documentation requirements, and general procedures.

The during phase focuses on completing the application itself. This is facilitated through a minimalistic, form-based interface that reduces complexity and helps users focus on the most important steps. The form is based on the official DiBK paper application form but has been adapted in a digital format. Several sections are enhanced or replaced using KartAi technologies such as TiltaksAiD and CADAiD, which increase the accuracy and completeness of the user's input.

To assist users throughout both the before and during phases, the application also includes an integrated chatbot, which acts as a virtual advisor. Available either on a separate page or as a floating chat bubble, the chatbot can help users with questions related to building laws, regulations, definitions, and general application guidance. TiltaksAiD is embedded directly in the chatbot interface, allowing users to access a map of their property and draw in the planned location and dimensions of their project. This interactive element supports a more dynamic and user-friendly application experience. The chatbot also retains conversation context, enabling ongoing assistance throughout the session.

The system is currently focused on the user-facing side of the application process. For this reason, the after phase, which primarily involves case worker tasks such as processing the application and updating its status, is only partially implemented. After submitting an application, the user is presented with a hardcoded interface that represents the confirmation and status update page. This lays the groundwork for potential future expansion on the case worker side.

10.2 Foundation for Further Development

The system is developed as a Proof of Concept and forms part of a larger research initiative. While not a finished product, it provides a strong foundation for future development in multiple areas. One of the key contributions is the establishment of functional architecture, where various components, such as the form-based application, chatbot integration, and embedded KartAi technologies, interact seamlessly to support the user throughout the building application process.

To support further work, basic documentation has been created. This includes in-code comments, a read.me file, and supplementary flowcharts and architectural diagrams designed to help new developers understand the system's structure and logic.

This prototype can serve as a starting point for developing a final version intended for deployment on the official platform of Kristiansand Municipality.

10.3 Future Plans

While the project the group has delivered a functional application with all the core concepts and ideas produced during the early design and development stages, there remain several functions and improvements that the group did not have enough time to implement. To support future iterations of this project, this subchapter will describe the group's ideas for further development in the overarching KartAi project.

One consideration the group discussed in the initial stages of the project was the possibility of the application catering to both the applicants and case workers. But as established, the group decided to focus on the applicants as the primary users due to time constraints. An implementation of the case worker's side of the application would be a valuable extension by

allowing the applicants and case workers to exist within the same system, making the workflows more efficient.

In addition to this, the group initially planned to implement, was “Before” and “After” input fields and categorisation, within the CADAiD step in the building application process. This feature was designed to facilitate the user’s ability to submit the current state of their property, as well as the proposed changes upon the project’s completion. This idea came as a suggestion from a building case worker, who highlighted that in many cases, it was difficult to distinguish between the current and proposed plans for a property.

Another feature was to implement the KartAi tool, ArkivGPT, an AI model that provides users with access to previous building projects associated with their property. Additionally, an examination of the property using DOK-analysis to determine factors affecting the possibility to build on the property would greatly inform the residents of future building opportunities. The plan was to have these technologies featured on the “Min Eiendom (My Property)” page, where they could gain insight as to what types of projects may be approved for their future building application.

In addition, the AI chatbot developed in this project has the potential for major improvements. As the KartAi project continues to develop, to combine all the existing chatbots in the KartAi project into a single, integrated chatbot. Currently, multiple independent chatbots exist in their own isolated project. Consolidating them into one cohesive chatbot would allow for the application of extensive knowledge when assisting the user.

Finally, there are smaller parts in the project which can be refined, such as inconsistencies between the front-end and the database. In some cases, the front-end is designed to display information that does not exist in the database. Additionally, the current step in the application process is not preserved when the user leaves the process and returns. Further, a full integration of “My Page (Min Side)” and a proper “Neighborhood Warning (Nabovarsel)” implementation in the application process, would greatly enhance the project’s UX and security in an official production. These features, considerations and refinements should be implemented in future iterations of this project.

11 Reflection

This chapter presents the group’s overall thoughts and reflections of this project in its entirety. It aims to evaluate the methods and decisions made throughout the development cycle and serves as an assessment of the current product’s quality compared to the previous Proof of Concept and security. Additionally, reflections on how prior experience, research and planning have impacted the project will be discussed.

11.1 Process and Methods

In the early stages of the project, we quickly identified certain knowledge and competence gaps that needed to be addressed. As mentioned in Chapter 3 and 4, we conducted extensive research and analyses to make up for the knowledge gap, particularly surrounding the needs of the applicants and case workers. This learning curve was eased through support from the stakeholder and case workers by allowing us to observe guidance sessions and gaining insight into the real problems applicants have.

Regarding the competence gap, the programming languages that the previous iteration was written in was partly unfamiliar to us, creating a steep learning curve. Therefore, ample time had to be given to both learn and understand the concepts of the programming languages. Moreover, the project required us to interpret and comprehend the work of the students who had developed on the previous Proof of Concept. This further contributed to a slow start in development, as substantial effort was needed to understand the various code components. At times, self-inflicted issues would occur, though these problems were usually resolved either through assistance from the technical advisor or through independent troubleshooting. In retrospect, we should have put aside more time to fully understand the functions and components of the Proof of Concept.

For competence gaps like these, clearer documentation of the codebase would have greatly sufficed in our understanding of the application and allowed us to focus on developing rather than deciphering the previous work. Because of this, we have committed to supplying sufficient documentation of our thoughts, ideas, and explanations of the code to support future iterations.

Our decision to conduct the research on both a technical level and in terms of user requirements and behavior, would turn out to be a favorable decision. On the technical aspect, we discussed the appropriate approach and architectural structure to ensure a maintainable and stable system. We made sure that any major changes or additions made to the application during development were discussed within the group to ensure a mutual understanding of the process. On the user-centered side, the research was frequently done through observation of guidance sessions and stakeholder discussions, resulting in a better and deeper understanding of the users' needs and challenges. This experience showed the importance of a two-sided research approach and was a good decision to make and resulted in a final product which was technically sound, while also aligning with user needs.

We decided to follow an agile Scrum methodology, as it aligned well with the dynamic nature of the project. Given the need for flexibility regarding new ideas, and input from the stakeholder, as described in Chapter 2, an agile approach allowed for quick adjustments and evaluation of process.

While there was no formal time limit to how long tasks needed to be done, we informally set out to complete most tasks within a week. Though, certain tasks would require more time due to unforeseen challenges. These delays would often affect subsequent tasks, resulting in lower-priority items being left unfinished. We underestimated just how much time certain tasks would take and should have set a formal time limit and made conservative estimations which would have allowed for better planning and flexibility surrounding development and task completion.

As this project has implementations of several KartAi tools which are still being developed, we had to make assumptions based on what we thought would be featured in these agents. For TiltaksAiD, we assumed that this technology would be integral into easing the building application process by being able to gather required legislative plans and documents. We discussed this possibility with the stakeholder as well, and it was confirmed that this was a feature they wanted implemented into the tool. In retrospect, instead of making assumptions, we should have had better communication with the project group in charge of the development of TiltaksAiD to clarify what exactly the tool does and will do in future iterations.

11.2 Product Quality

The project we have delivered, as an expanded version of the NTNU students' Proof of Concept, will inherit some qualities from the earlier work while also adding new ones.

For instance, the knowledge of the KartAi technologies that were implemented remains the same except for one. The previous PoC had a chatbot, which was further developed in the new version of the project. The difference between the old and new chatbot is of the differing purposes. The previous chatbot gave information about regulation plans, while the further developed chatbot is a tool that provides both information of regulation plans but also provides knowledge of what the user can potentially build on their property and give more user specified answers. Additionally, the KartAi tool TiltaksAiD was integrated into this project, a model the PoC did not have. Although the implementation of TiltaksAiD was requested by the stakeholder, the expansion of the established chatbot's knowledge was a decision made by us. We believe that this additional knowledge, as expounded upon earlier, will be a valuable tool in assisting the applicants in the long term.

Further, the design is arguably more user-friendly. Though design is subjective, the feedback we got from the stakeholder was abundantly positive. They claim they are very happy with the design and favor this project's UI over the previous PoC. We presented various designs to the stakeholder, and these designs have always been positively received, and small details we might have missed but were caught by the stakeholder were quickly resolved. This experience showed how keeping a constant dialogue between us and the stakeholder would lead to a satisfactory product.

In regard to security, we initially thought that this aspect would be one of the major implementations of the project. After discussing and evaluating the topic with the stakeholder, it was decided that security would be outside of the project scope, and therefore not included as a concern in this project. The reason for this was due to our capacity and ability as a group to safely and securely implement steadfast security in the project. If the project would be put to production and be accessible as an official solution for digital building applications, then security would have to be thoroughly implemented due to communal processes, privacy, personal information and sensitive data among others.

While all the concepts mentioned are just a few of the features that we did not have enough time and capacity to implement, this experience overall has taught us not to underestimate the

time we have been given. Meetings, tasks, discussions, among others, were instances where much more time was taken than we had initially thought. As we reflect on this in the aftermath, we will take this into consideration when working on future projects.

12 Conclusion

This bachelor project has successfully demonstrated how AI technologies from the KartAi initiative can be practically applied to improve the digital building application process. By transforming an earlier Proof of Concept into a functional and user-centered prototype, the project addressed the primary challenges outlined in the problem statement, particularly the complexity of regulations, frequent submission errors, and prolonged case processing times.

The solution developed supports applicants in submitting complete and accurate one-step building applications by integrating three key AI tools: a contextual chatbot for regulatory guidance, CADAiD for document validation, and TiltaksAiD for spatial zoning analysis. This directly answers the first research question, showing how AI and data-driven tools can streamline and digitize the application process. The guided digital flow and proactive chatbot support demonstrate how residents can intuitively submit complete applications, while the structured outputs and reduced error rates support case workers in processing applications more efficiently.

All of the project's core objectives were met: we enhanced the original Proof of Concept, improved the user interface and experience, validated the functionality of integrated AI models, and delivered a working prototype along with documentation to guide future development. Although the scope was limited by time and external dependencies, particularly the evolving nature of KartAi technologies, the project lays a strong technical and conceptual foundation for future iterations.

In addition to fulfilling the academic goals of the bachelor course, this project illustrates the real-world potential of combining AI technologies with user-centered design to modernize public services. Future work could include developing a complementary interface for case workers, extending chatbot capabilities, and integrating archival and real-time municipal data. With further refinement, this solution could become a key enabler of more efficient, transparent, and user-friendly building application processes. This project illustrates not only the feasibility of such an approach but also the value of combining AI technologies with thoughtful, inclusive design in public-sector innovation.

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Appendices

Appendix 1 - Statement from Kristiansand Municipality

Atlas har fra første dag levert over forventning. De har konsekvent kommet godt forberedt, og satte seg raskt inn i komplekse fagfelt hos plan og bygg. Vi er spesielt imponerte over deres proaktive nysgjerrighet, som har vært avgjørende for innhenting av vesentlig informasjon for utviklingen av en løsning som oppfyller våre behov. Forståelsen deres av fagfeltet ble ytterligere styrket av en vel utført oppgave om digitalisering i offentlig sektor. De har hatt fast kontorplass en gang i uken samt utvidet ved behov, noe de ikke nølte med å ta i bruk. Gjennom en tidlig grovanalyse, ukentlige statusmøter og nær kontakt med oppdragsgiver, avgrenset de en bred og lite definert oppgave til et gjennomførbart prosjekt som legger et solid grunnlag for videre utvikling. De har tatt alle utfordringer på strak arm og vært et positivt innslag i kontorlandskapet. Vi takker for et fantastisk godt samarbeid med et resultat vi er stolte av å vise frem, og ønsker studentene lykke til videre.

Vennlig hilsen,

Kristiansand kommune



Sébastien Thune

KI-rådgiver | Kristiansand Kommune - Plan og bygg, Geografisk Informasjon

Appendix 1.1 Statement from Building Case Worker

Tilbakemelding KartAI-prosjektet - Atlas

Anna Maria, Maria og Nicoletta, som gruppen Atlas, har jobbet iherdig med prosjektet om utstillingsvindu i forbindelse med sin Bachelor-oppgave og KartAI-prosjektet. De satt seg fort inn i oppgaven, og har imponert med god struktur og planlegging til alle møter. De har hatt en klar rollefordeling innad i gruppen, noe som har vært fint for oss andre i prosjektet å forholde oss til. Som byggesaksbehandler er jeg imponert over hvor godt de har satt seg inn i lover og praksis rundt byggesaksbehandling. De har gjort research via selvstudier, spurt om hjelp og vært med på flere veiledningsmøter som byggesaksbehandlerne hos oss har med innbyggerne i kommunen. På denne måten har de fått god innsikt i både hva innbyggerne lurte på, og hva vi som byggesaksbehandlere trenger for å behandle byggesøknader. De har tatt til seg innspill på en god måte gjennom hele prosessen, noe jeg som saksbehandler setter stor pris på. Utstillingsvinduet og det forslaget til innbyggerdrevet løsning de har presentert vil kunne hjelpe mange innbyggere i alle landets kommuner med å sende inn komplette søknader. Løsningen er intuitiv, gir god informasjon og veiledning. Det er flere Bachelor-studenter med i KartAI-prosjektet, og Atlas har klart å inkorporere deres løsninger i sitt eget prosjekt på en veldig god måte. Dette viser gode samarbeidsevner. Atlas har kun vært tre personer i gruppa, og jeg er derfor imponert over arbeidskapasiteten deres og hva de har fått til på dette snaue halve året. De fremstår som veldig faglig sterke og nysgjerrige.

Med vennlig hilsen

Mari Gunnerud Haus,

Fagleder for veiledningstjenestene og 3-ukers byggesaker, og jurist

Appendix 2 – Group Evaluation

Anna Maria Dang:

I was the group leader and UX designer for the project. My responsibilities included coordinating the project, organizing meetings, and ensuring smooth communication both within the team and with external stakeholders. I planned and facilitated meetings, followed up on tasks, and helped keep the project on track.

As a UX designer, I worked on the overall user experience, focusing on how users interact with the chatbot and the application. I created the Figma prototype, which served as the foundation for the frontend development, and made sure the design was user-friendly, consistent, and aligned with the project's goals.

In addition to my coordination and design work, I also had the primary responsibility in writing the final report and helped ensure the documentation reflected both the process and the outcome of our work.

Maria Anh Thu Hoang:

In this project, I was the front-end developer, responsible for building the user interface and ensuring a smooth and consistent user experience. I followed the Figma prototype closely to make sure the design was implemented accurately and reflected the intended look and feel of the application.

My focus was on creating a clean and intuitive interface that made it easy for users to interact with the chatbot and navigate through the system. I worked closely with the back-end developer to integrate the chatbot functionality and ensure data flowed correctly between the front-end and back-end. My contributions also include carrying out analyses and writing of the thesis.

Throughout the project, I paid attention to detail, responded to feedback, and made design adjustments where needed. After the primary development period, I also had the task of refactoring the front-end pages and components, to ensure good programming practices and making sure the application ran smoothly.

Nicoleta Pavelescu:

As the primary back-end developer on the project, my main responsibilities involved designing and implementing back-end functionality to support the chatbot and associated services. This included setting up API endpoints, integrating AI models such as CADAiD and TiltaksAiD, and ensuring seamless communication between the back-end and front-end components.

I was actively involved in the development of the AI agent, contributing to its logic and interaction flow. I supported its integration with LangChain's React Agent and enabled tool usage through well-structured backend services. I also participated in implementing a testing strategy aligned with agile development practices, where I used the LangSmith platform to trace the chatbot's behavior, analyze token usage, identify errors, and optimize performance. Additionally, my contributions include writing the thesis and doing analyses.

Throughout the project, I contributed to both manual and assisted testing, developed back-end support for tool invocation, and helped debug issues by analyzing logs and agent traces. My work ensured the system was both technically robust and responsive to user inputs, laying a solid foundation for further development and scalability.

Appendix 3 – Group Contract

Gruppekontrakt

Vi, gruppe 13 som består av medlemmene: Anna M. Dang, Maria A. T. Hoang, og Nicoleta Pavelescu. Skal jobbe sammen om gruppearbeid i det sjette og siste semesteret i bachelorstudiet IT og informasjonssystemer. Vi skal samarbeide i følgende emne: IS-304 og IS-305.

Bakgrunn og motivasjon

Vi som gruppe ønsker å lære om og mestre læremålene som er oppgitt på Canvas, samt å oppnå best mulig resultat i emnene IS-304 og IS-305. Gruppekontrakten ble opprettet og gått igjennom tirsdag 10.01.2025, der gruppe medlemmene ble enige om retningslinjer for det videre arbeidet innenfor det sjette semesteret i studiet IT og informasjonssystemer.

Mål: Samarbeide for å fullføre bachelorstudiet med best mulig karakter og øke kunnskapsnivåene våre.

Gruppens regler:

- Alle avgjørelser skal gjøres i fellesskap.
- Ved uenighet skal avstemming benyttes.
- Alt produsert materiale skal ha backup, og korrekturleses av minst et annet gruppe medlem.
- Informasjon til gruppe medlemmer skal sendes på Discord.
- Møteplikt på alle møter, unntak ved gyldig fravær (f.eks. sykdom) med melding av fravær.
- Hvert enkelt gruppe medlem plikter å fullføre tildelte oppgaver til oppsatt tid. Dersom dette ikke lar seg gjøre skal gruppe leder informeres så tidlig som mulig. Gjentatte overtredelser kan medføre at fagpersonell blir informert, noe som kan skape konsekvenser for vedrørende gruppe medlem.

Bestemmelser om fremmøte og samarbeid

Jeg forplikter meg overfor gruppen min, til at jeg vil:

- Følge forelesninger og/eller følge pensum, og delta i gruppearbeid, samlinger og øvelser, be om/ta imot veiledning, gjøre selvstudium av litteratur og delta aktivt i gruppearbeid med feltstudier.

- Respektere kravet om minst 80% frammøte og også oppgi grunn hvis jeg er forhindret. Fravær gjelder om det ikke er noe kommunikasjon/innsats mellom enkelte gruppemedlem og gruppen. Unngå fravær i henhold til gruppas kommuniserte normer. Gi beskjed om fravær om dette skulle hende.
- Overholde frister og levere oppgaver som publiseres på Canvas/Inpera, innen fristene. Sørge for at arbeid fordeles jevnt mellom gruppemedlemmer, men samtidig utnytte hver enkelt students spesielle ferdigheter og bakgrunn.

Andre bestemmelser

Ansvarsfordeling som følger:

Gruppeleder: Anna Maria Dang

- Gruppeleder har ansvar for:
 - Få i gang diskusjoner.
 - Å ha kontroll over møter (når/hvor) og informere alle gruppemedlemmer.
 - Delegere oppgaver til alle gruppemedlemmer.
 - Sørge for at alle blir sett og er aktiv i gruppediskusjoner.
 - Å ha kontroll og informere om fremdrift av oppgaver/prosjekter.
 - Skrive notater fra møtene.
 - Ha kontakt med forelesere, veileder og kontaktpersoner for bachelorprosjektet og i andre oppgaver om det er behov.

Signatur:

Dato: 10.01.2025

Anna Maria Dang

Nicola Parkesen

Maria Anh Thu Hoang

Appendix 4 – Overview of each sprint

Sprint	Dato begynt	Dato slutt	Fase	Mål	Viktig dato!	Notater
Pre-sprint	02/01/25	16/01/25	Undersøkelse og Analyse	Forstå hva oppgaven er og hva vi skal gjøre.	Forelesning oppstart: 8. januar <u>KartAi</u> oppstart: 10. januar	
Sprint 1	17/01/25	24/01/25	Undersøkelse og Analyse og Ide	Ha flere forslag til design basert på undersøkelse, analyse og ide	Workshop IS-305: 24. januar	
Sprint 2	25/01/25	31/01/25	Ide og Design	Velg et design og begynn å utvikle	Kick-off <u>KartAi</u>: 28. januar	
Sprint 3	01/02/25	07/02/25	Design + Prototype	Prototype klar til brukertesting?		
Sprint 4 UKE 7	08/02/25	14/02/25	Prototype (Brukertesting ?)	Utfør brukertesting for å se om det funker bedre enn eksisterende	Status 1: 12./13. februar	Lag en prototype som kan bli testet av VGS elever.
Sprint 5	15/02/25	21/02/25	Prototype + Utvikling	Begynn å utvikle basert på design prototypen og brukertesten		Kvalitetssikring underveis.

Sprint 6	22/02/25	28/02/25	Utvikling	Se om det er noe som må endres.		Kvalitetssikring underveis.
Sprint 7	01/03/25	07/03/25	Utvikling	Fokus på å lage webapplikasjon for digital byggesøknad og å lære for chatbotten.		Kvalitetssikring underveis.
Sprint 8	08/03/25	14/03/25	Utvikling	Fortsette med webapplikasjon og øke kompetanse. Begynn å skrive på rapporten.	Voluntary supervision: 5. mars	Kvalitetssikring underveis.
Sprint 9	15/03/25	21/03/25	Utvikling	Fokus på webapplikasjonen og chatbot.	Status 2: 12./13. mars	Kvalitetssikring underveis.
Sprint 10	22/03/25	28/03/25	Utvikling	Fokus på webapplikasjonen og chatbot.		Kvalitetssikring underveis.
Sprint 11	29/03/25	04/04/25	Utvikling	Fokus på webapplikasjonen og chatbot.	Voluntary supervision: 26. mars	Kvalitetssikring underveis.
Sprint 12	05/04/25	11/04/25	Utvikling	Fokus på webapplikasjonen og chatbot.		Kvalitetssikring underveis.
Sprint 13	12/04/25	18/04/25	Utvikling	Fokus på webapplikasjonen		Kvalitetssikring underveis.

				og chatbot.		
Sprint 14	19/04/25	25/04/25	Utvikling	Fokus på webapplikasjonen og chatbot. Skrive i rapporten.	Final report IS-305: 24. april kl 15:00	Kvalitetssikring underveis.
Sprint 15	26/04/25	02/05/25	Utvikling	Fokus på webapplikasjonen og chatbot. Skrive i rapporten.	Konferanse med presentasjon og review: 29. april	Kvalitetssikring underveis.
Sprint 16	03/05/25	09/05/25	Utvikling	Fokus på webapplikasjonen og chatbot. Skrive i rapporten.	Code freeze: 9. mai (senest senest 14. mai)	Kvalitetssikring underveis.
Sprint 17	10/05/25	16/05/25	Finalisering av rapporten og utvikling	Ferdigstille rapporten og sende den inn.	Innlevering: 16. mai kl. 12	
Sprint 18	17/05/25	02/06/25	Finalisering av utvikling	Fikse at Azure VM funker. Prøve å utføre brukertester. Ferdigstille dokumentasjon av prosjektet for hand off.		

Appendix 5 – The Backlog

	Order	ID	Title	Assigned To	State	Tags
1		119	📌 Chatbot Veileder	Nicoleta Pavel...	● To Do	
		120	✅ Arkitektur	Nicoleta Pavel...	● Done	
		121	✅ Kartlegge funksjonalitet til chatbotten (hva kan den gjør...	Nicoleta Pavel...	● Done	
		122	✅ Interagere med TiltaksAiD (kartet)	Nicoleta Pavel...	● Done	
		123	✅ Lagre objekter fra kartet		● To Do	
		124	✅ Registrere objekter fra kartet	Nicoleta Pavel...	● Done	
		125	✅ Lagre samtalen med søker	Nicoleta Pavel...	● Done	
		126	✅ Hente adresse til søkeren	Nicoleta Pavel...	● Done	
		127	✅ Oppsummere reguleringsplaner		● To Do	
		128	✅ Lage mockup data! (Alt trenger ikke å være på plass til s...	Nicoleta Pavel...	● Done	
		139	✅ Funksjonalitet: Kartlegge hva som er kjappast å gjøre sli...	Nicoleta Pavel...	● Doing	
		140	✅ Hvis man ikke er logget inn, få melding fra chatbotten: s...		● To Do	
		141	✅ Feilmeldinger i selve chatten, og info om hva man har te...	Nicoleta Pavel...	● Done	
		142	✅ Feilmelding om at chatbotten ikke fungerer i chatten, ikk...	Maria Anh Thu...	● Done	
		143	✅ Vise til riktig veiviser	Nicoleta Pavel...	● Done	
		153	✅ Sette opp database tabeller for søknad og nabovarsel	Nicoleta Pavel...	● Done	
		160	✅ Implementere metode for at chatbotten skal legge til inf...		● To Do	
		161	✅ Metode for at chatbotten skal oppdatere info i databasen		● To Do	
		162	✅ metode for at chatbotten skal slette info fra databasen		● To Do	
		195	✅ Embedde reguleringsplan (kommuneplanen og regulerin...	Nicoleta Pavel...	● Done	
		196	✅ Lage workflow	Nicoleta Pavel...	● Done	
		197	✅ Implementere search_tool	Nicoleta Pavel...	● Done	
		198	✅ implementere tool for rag	Nicoleta Pavel...	● Done	
2		199	✅ implementere tool for api kall		● To Do	
		200	✅ implementere ReAct agent	Nicoleta Pavel...	● Done	
		99	📌 Issues		● To Do	
		100	✅ Noen bilder i cadaid blir ikke evaluert	Nicoleta Pavel...	● Done	
		101	✅ Fjerne userDocuments tabellen	Nicoleta Pavel...	● Done	
		102	✅ Lagre bilde i database kun når den er godkjent av Cadaid	Nicoleta Pavel...	● Done	
		103	✅ Koble fjern document knappen mot database	Nicoleta Pavel...	● Done	
		104	✅ Vise dokumenter i form av bilder istedenfor en liste av d...	Nicoleta Pavel...	● Done	
		105	✅ Implementere testere for api	Nicoleta Pavel...	● Doing	
		106	✅ Lese oss opp på dok-analyse dokumentasjonen	Nicoleta Pavel...	● Done	

	109	✓ Gjør det mulig å forstørre bilde i CadAld	Maria Anh Thu... ● Done
	146	✓ Fjerne kode vi ikke bruker	Maria Anh Thu... ● Doing
	190	✓ Hente applicationID i alle komponenter	Maria Anh Thu... ● Done
	191	✓ Hente informasjon fra database for dispensasjon/Nabov...	Maria Anh Thu... ● Done
	192	✓ Passe på og lagre all informasjonen fra søknad i databas...	Maria Anh Thu... ● Done
	193	✓ Fjerne komponenter som ikke brukes (sjekk beskrivelse)	Maria Anh Thu... ● Done
3	65	📁 Komponenter	● To Do
	66	✓ Progressbar	Maria Anh Thu... ● Done
	67	✓ Buttons	Maria Anh Thu... ● Done
	68	✓ Sjekkliste	Maria Anh Thu... ● Done
	69	✓ Sidebar	Maria Anh Thu... ● Done
	71	✓ Sjekkliste (oversikt)	Maria Anh Thu... ● Done
	227	✓ min oversikt	Maria Anh Thu... ● Done
	228	✓ før du søker	Maria Anh Thu... ● Done
	229	✓ min eiendom	Maria Anh Thu... ● Done
	230	✓ mine søknader	Maria Anh Thu... ● Done
	231	✓ mine dokumenter	Maria Anh Thu... ● Done
	232	✓ meldinger page	Maria Anh Thu... ● Done
	233	✓ tiltaksaid	Maria Anh Thu... ● Done
	234	✓ chatbot	Nicoleta Pavel... ● Done
	235	✓ small chatbot	Maria Anh Thu... ● Done
	236	✓ to do list in application process	Maria Anh Thu... ● Done
4	35	📁 Sette opp Figma	Anna Maria Da... ● Doing
	36	✓ Finn color scheme	Anna Maria Da... ● Done
	37	✓ Lag felles komponenter	Anna Maria Da... ● Done
	76	✓ Fiks wireframe	Anna Maria Da... ● Done
	77	✓ Fiks analyser	Anna Maria Da... ● Done
	82	✓ Lage prototypen	Anna Maria Da... ● Done
5	203	📁 issues	● To Do
	204	✓ Fikse eksisterende søknader på min-oversikt når du ikke...	Maria Anh Thu... ● To Do
6	201	📁 Refactoring	● To Do
	202	✓ Component PlanChatAtlas.tsx	Maria Anh Thu... ● Doing
	205	✓ Application Process	Maria Anh Thu... ● To Do

206	✓ Sidebar items	Maria Anh Thu... ● To Do
207	✓ TypeScript errors	● To Do
208	✓ My overview	Maria Anh Thu... ● To Do
209	✓ før du søker	Maria Anh Thu... ● To Do
210	✓ min eiendom	Maria Anh Thu... ● To Do
211	✓ mine søknader	Maria Anh Thu... ● To Do
212	✓ mine dokumenter	Maria Anh Thu... ● To Do
213	✓ meldinger page	Maria Anh Thu... ● To Do
214	✓ TypeScript warnings	● To Do
215	✓ Remove note comments	● To Do
216	✓ Add xml documentation	● To Do
217	✓ Minor front-end components	Maria Anh Thu... ● To Do
218	✓ TiltaksAiD	Nicoleta Pavel... ● To Do
219	✓ Chatbot	Nicoleta Pavel... ● To Do
220	✓ Sjekkliste sidebar	Maria Anh Thu... ● To Do
221	✓ Progressbar	Maria Anh Thu... ● To Do
222	✓ Application Process to do tasks	Maria Anh Thu... ● To Do
223	✓ Duplicate application process steps across application t...	Maria Anh Thu... ● To Do
224	✓ delete unused components	● To Do

Appendix 6 – Full Risk Matrix

Risiko	Konsekvens	Sannsynlighet	RISIKO	Tiltak for at det ikke skal skje	Tiltak, redusere skade
Gruppemedlemmer mister motivasjon. (very sad)	4	2	8	Hjelpe hverandre, være sosiale, ta pauser, <u>good</u> vibes og <u>energy</u> .	Ta oppgaver litt om gangen. Færre og enklere oppgaver for å bygge seg opp igjen.
Noen gruppemedlemmer taper til sykdom. (sad)	4	3	12	Ta vare på <u>seg</u> selv.	Ta pauser.
Ikke tilfredsstilte <u>stakeholder</u> .	4	1	4	Ha tett kommunikasjon om hva stakeholder vil ha (ukentlige møter).	Prioritere det som er viktigst av det stakeholder vil ha. Ha tettere dialog med stakeholder.
Ikke ha de nødvendige ferdighetene for programmering.	4	3	12	Self-study, sette <u>seg</u> inn i koden tidlig. <u>Være</u> forberedt.	Peer-review av koden før det blir pushet opp på Github.
Gruppen er ikke i enighet om deltakelsen i prosjektet.	2	1	2	God kommunikasjon. Alle gruppemedlemmene er med å bestemme.	Gruppmøte for å avklare deltakelse.
Gruppen er ikke enige om parametrene i prosjektet.	3	2	6	Gruppen avklarer med hverandre og stakeholder om parametrene.	Holde flere møter for å forbedre informasjonsflyten og for å kunne ta samlet beslutninger.
Kommunikasjonen faller sammen mellom gruppen og <u>stakeholder</u> .	4	1	4	Faste planlagte møter.	Ta initiativ/ initialisere møter. Gi beskjed til emneansvarlig.
Tilgang til KartAi modeller.	4	1	4	Tett dialog med stakeholder og AI-provider. Ha god kontakt med stakeholder, ikke vær redd for å spørre spørsmål.	Ta kontakt med stakeholder i etterkant.
Noen gruppemedlemmer	1	4	4	Gruppen deler kunnskap de har etter hvert	Lære hverandre i etterkant.

har mer kunnskap enn andre. Hvert gruppemedlem spesialiserer seg i et fagfelt.				Daily Scrum. Dele gruppen i roller.	
Nøkkelpersonell er kanskje ikke tilgjengelig ved behov.	3	2	6	Planlegge godt i tid. Ta kontakt så tidlig som mulig.	Kontakte andre egnede personer.
Uklare krav.	4	2	8	Kommunisere med stakeholder. Ha tett dialog med stakeholder i løpet av hele prosjektets levetid.	Møte med gruppen, stakeholder, og veileder for å avklare krav.
Har ikke nok tid til å gjøre alle oppgavene som har blitt satt.	4	2	8	Prioritere oppgaver som er viktigst å gjennomføre. Dele ansvar mellom gruppemedlemmer .	Innsnevre prosjektets omfang.
Dårlig Git-hygiene.	3	3	9	Sette tydelige retningslinjer for strukturen av Github-en .	Kontroll underveis, om det blir for dårlig, finn en løsning for eks. en hovedansvarlig.
Feilplassert oppmerksomhet.	4	3	12	Kommunisere med kommunen for tilbakemelding. Fokuser på problemstillingen til prosjektet.	Møte med gruppen, stakeholder, og veileder for å avklare krav.

Appendix 7 – Estimated Time Distribution by Activity

Each group member worked on average +/- 37,5 hours each week on the project. Here is an estimate of the time distribution by activity. Note that it is blocked into what the group considers as the project's key activities. See the backlog on Appendix 5 for more detailed tasks.

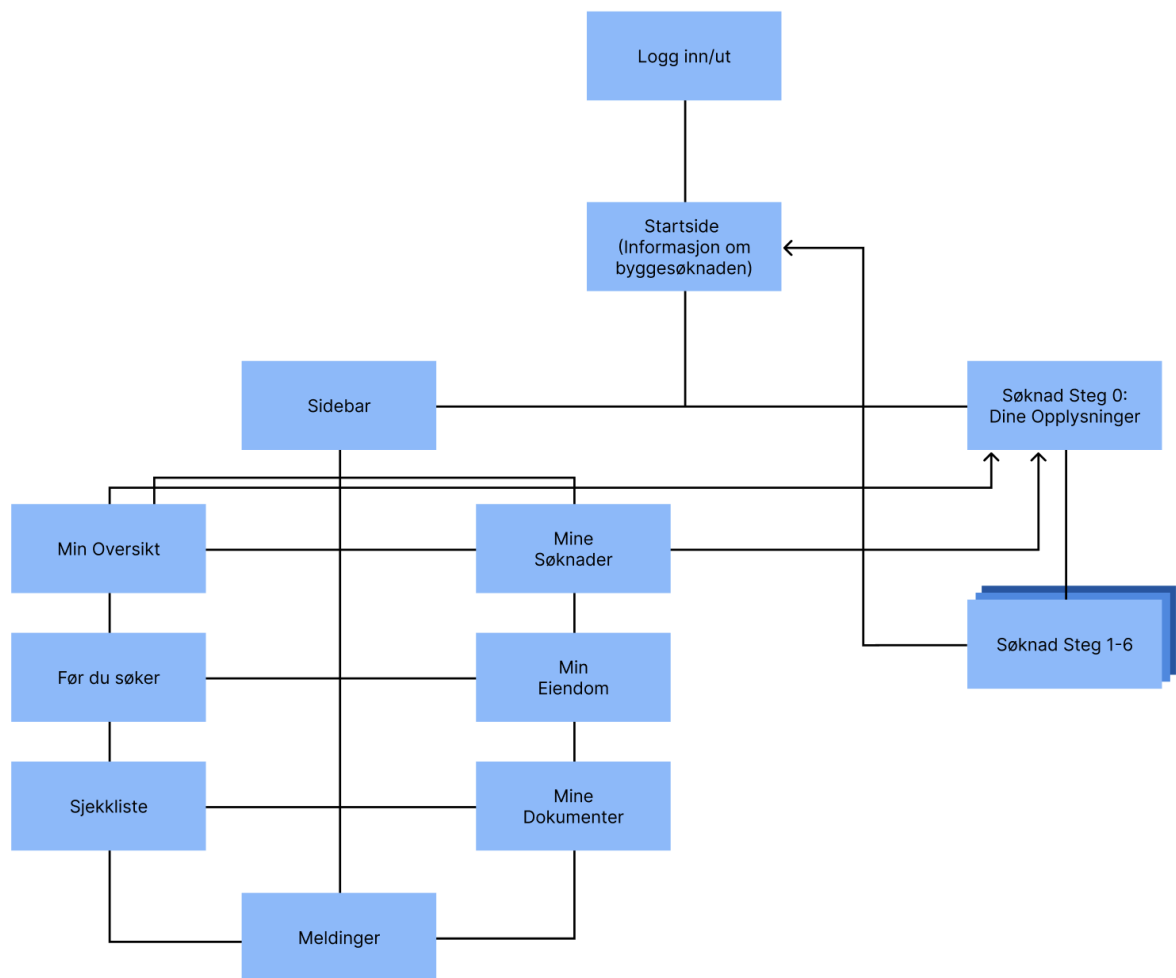
Activity	Description	Estimated Time	Real Time
Project Management	Time used on project management, for example: planning, decision making, quality assurance, etc.	100 hours	Ca. 150 hours
Background Research	Times used on reading and learning for the project.	200 hours	Ca. 350 hours
Figma (Design/UX/UI)	Time used on the design process and making the wireframes, prototype and the initial thought process.	200 hours	Ca. 225 hours
Front-end development	Time used on developing the front-end of the application.	300 hours	Ca. 370 hours
Back-end development	Time used on developing the	400 hours	Ca. 500 hours

	back-end of the application.		
Idea Pitch	Time used on the idea pitch of the chatbot.	2 hours	5 hours
Testing	Time used to test the application.	10 hours	Ca. 50 hours
Code Refactoring	Time used to refactor the code of the application.	70 hours	Ca. 125 hours
Report Writing	Time used to write the report.	50 hours	Ca. 100 hours
Other	Time used on other parts of the project.	100 hours	Ca. 50 hours

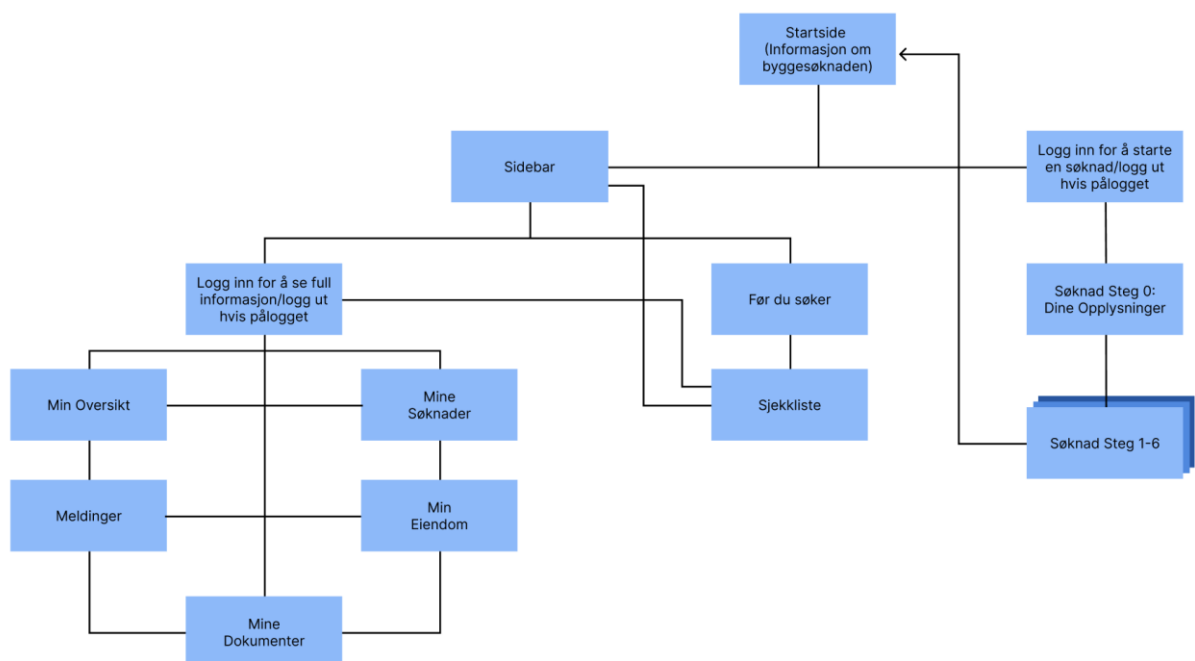
Total = 1925 hours / 3 = 641.67 hours (total per person) / 17 = 37.7 hours (per person per week). The time given is a realistic estimate of the time the group spent in total on the project as we did not formally log hours, thus the real time may differ from what was spent.

Appendix 8 – Navigation Map

Navigation Map whilst Logged In:



Navigation Map not Logged In:



Appendix 9 – Link to Figma

To access the Figma file which shows all the workspace, use the password “Atlas”.

Disclaimer: Not all the work was finished due to time constraints and unforeseen challenges, but it was used as the foundation for the web application.

Link: <https://www.figma.com/design/3g2mACIqFdpsI7vonAoqOY/KartAi---Utstillingsvindu-2.0-Laget-av--Atlas?node-id=0-1&t=BxJRLvvTkKttUxl6-1>

Appendix 10 – Idea Pitch Presentation

Link to the chatbot idea pitch presentation:

<https://www.figma.com/slides/KtPPATJOkYGnSSnwcaraAe/Pitch-Chatbot-Veileder?node-id=1-42&t=JFHW6kGxdPx16Ivc-1>

Appendix 11 – Observation Session Notes

Dato: 11/02

Nabokonflikt

Saksbehandler notater:

- Viser eiendommen på kart
- Har du vært på nettsidene? søknadspliktig eller ikke
- Gå inn på nettsiden i arkivene
 - Ikke godkjent er ikke godkjent
- Foreslår om å be om innsyn
 - Viser søker til hvor på **innsyn**
 - Be søkeren om å gå inn på byggesaksarkivet
 - Bedre måte å lese dokumentet på enn på mobil
 - Problemer med å komme inn -> be om innsyn
 - Finn en enhet man kan ordentlig få opp alle dokumentene på
- Problemer med å finne ut hva som er godkjent?
 - Gå gjennom kommunens nettsider
 - Området til søker er uregulert, kan komme en dispensasjon hvis det er mange tiltak, ha gode fagfolk rundt seg
 - Mye vekt på byggesaksarkivet å forholde seg til det
 - Forholde seg til de tegningene som er godkjent fra før

Søker notater:

- Spørsmål:
 - Nabokonflikt
 - Bygde en veranda, ikke søkt om, ikke lov
 - Ligger for nærme nabogrensen
 - I 1992, fikk faren godkjent verandaen
 - I dag, verandaen skulle bygges ut, besøk av en arkitekt, naboen sier nei, dersom det når nabogrensen
 - Er det godkjent eller søkt om bod under verandaen?
 - Søker kommer ikke inn på link for arkiv
 - Ingen dokumenter kommer opp
 - Saksbehandler: det er en annen link man må inn på for tidligere år.
 - Telefonen er for liten
 - Problemer med å komme inn -> be om innsyn

Ulovlig oppfølging sak

- Søker hadde bygd inn på nabogrensen
- Innsyn for å få dokumentet
- Kart hvor man kan se tiltak for TiltaksAiD
- DIBK har hjelpemidler men ikke for kartet
- For det meste veiledning til søknadsprosessen
- Hvordan søker du og slik søker du

Dato: 14/02

Vil bygge terrasse

Saksbehandler notater:

- Forskjellige typer terrasse
 - Noen trengs å søke og ikke søke
 - Er platting tatt vekk?
 - Sjekker adressen til søker
- Snakker om målinger og avstand til nabo
 - Kan ikke bygge inntil 4 m til nabogrense
 - Nærmere enn det kreves samtykke fra nabo
 - Nabovarsel != **nabosamtykke**
 - Om nabo ikke samtykker, må søker søke om dispensasjon
 - Om nabo samtykker, trenger ikke dispensasjon, kommunen tar en vurdering
 - Ser på planavgrensning. En del av huset er innenfor en planavgrensning, men den andre delen ligger innenfor kommuneplanen, kan komplisere søknaden
 - > 50 kvm på det som er i kommuneplanen, må søke om dispensasjon, om > 50 kvm, i det regulerte, må forholde seg til reguleringsplanen
 - Til søker: Må melde inn nabo om at naboen er over nabogrensen
 - Viser på kommunens nettside

Søker notater:

- Vil bygge terrasse
 - Spørsmål: Hva er veien videre?
 - Bygges inn til boligen
 - Sprengt fjell oi, mistet uteplass pga det, faller det vekk?
 - Platting er tatt vekk, støpt platting
 - Vil bygge UT terrassen inntil 5 meter
 - Terrasse på stolper, 100 kvm
 - Kan ikke stå ansvarlig for det
 - Må søke som selvbygger (ansvarsrett), kan søke om ansvar om det, kan få en snekker til å søke om det
 - > 50 kvm må søke om selvbygger
 - Snekker gjør søknaden
 - Saksbehandler sier:
 - Må tegne planen på et situasjonskart, printe ut
- Spørsmål om naboen
 - Naboen har satt opp en fyllingsbod som står utenfor grensen

- I utgangspunktet ikke lov. Han er over grensen
- Kan nekte fyllingsboden
- Minimum 1 meter fra nabogrensen
- Nabo kjører inn på han sin tomt med gravemaskin og sånt
- Prengt fjell, noen lepper endte opp på naboen sin eiendom, kommunen ordna det
- Terrenginngrep er også avhengig av nabogrensen
- Nabo fikk aldri nabovarsel da søker sprengte, det var før han tok over eiendommen. Nabo klaget til kommunen.

Dato: 21/02

Utvide garasje

Topdalsveien 263

- Uregulert
- Det røde er faresone

farger

- planregister
- arealplankart

Saksbehandler notater:

- 2019 annet arkiv
- Viser saken*
 - Sjekker hva som står i vedtaket (tilbygg)
 - Sjekker dispensasjon dokumentet manuelt
 - Søker har fått dispensasjonen
- Spør om søker har vært i historisk arkiv på krs side
- Må egentlig søke om ferdigattest*
- "Melding om tiltak uten søknadsplikt"
 - Kan fylle inn et skjema, dibk
- Sjekker irl view av eiendommen
- Skal egentlig bare finne ut av hva søkeren må gjøre
- Har interaktiv veileder (klikke på ting)
- Kan være søkere som spør om konkrete bygg, da vil en saksbehandler ha bedre svar enn en veileder

Søker notater:

- Vil se hva som er fritatt fra å søke (under 50 kvm)
 - Frittliggende bygg under 50 kvm, og holder seg unna 1 meter fra nabogrensen
- Vært på direktoratet og kommunen sin side, mange sider
- Sett på reguleringsplan fra 1970 - står ingenting, ingenting nyttig
- Ikke meldt inn om at noe er bygget pga corona, men den er godkjent før det*

- Søkte om dispensasjon
- Må sende inn skjema, ingen faktura
- Sendte inn kart av hva de skulle gjøre for utvidelsen av garasjen
- Snekker
- Blir mye å bruke konsulenter for å finne ut av alt dette
- Søker lur på hvordan man kan finne ut av hva naboen har gjort/oppført
 - Naboen har bygd på nabogrensen
 - Byggesaksarkiv, min side
- Sendte nabovarsel om tilbygget, nå er det bare beskjed om utvidelse av garasjen
- Navigere gjennom skjema til dibk. Spør om ting til den
- Går gjennom veiviser bygg uten å søke dibk
 - Søker gikk gjennom det før
- Eiendommen er regulert
- Hvor langt vekk fra veien er garasjen
- Kan hende søker velger feil veiviser (tilbygg eller bygg uten å søke)
- Konklusjon: trenger ikke å søke

Dato: 28/02

Dispensasjon

Saksbehandler notater:

- Fasadeendring, søknadspliktig
- Taket, mange tiltak, et og et tiltak om gangen er så å si ikke søknadspliktig, men når man tar mange om gangen så er man mer sannsynlig til å være søknadspliktig
 - Endrer uttrykket i landskapet
- Søknad om mange tiltak vil være én vurdering
- Det må være gode grunner for å gjøre endringene
- Mange av de tidligere grunnene til endringene som har blitt gjort allerede, kan bli brukt til dette tilfellet
- Krav til tegninger
 - Slik søker du
 - 1:100 målestokk
 - Viktig at det er i målestokk, målsatt og at saksbehandler vet hva som skjer
 - Tegn inn på situasjonskart også
- Anbefaler å søke om pipa uavhengig (prioriteringer?)

- Pipa og taket først
- Pipa behandles som fasadeendring, ikke dispensasjon, kan behandles fort
- Anbefaler å søke om det som kan bli behandles fortest først
- Søker vil rive noe slitt mur, kan egentlig rive uten å søke
 - Med de tingene som er unntatt, så er det best å sende mail om det
- Saken må til parkvesenet for uttalelse
- Søker kunne ha løst problemet selv hvis de hadde fulgt sjekklisten
 - Har jo ingen forutsetninger
 - Kommer til veiledning fordi søker ikke kan det
- Har noe lignende kart i Danmark for å tegne inn og plassere i kartet, Arendal og Osl har prøve for det som er unntatt søknadsplikt
- Byggesak.ai
 - Føler man egentlig får nesten like svar i copilot
 - Vil at byggesak.ai skal bli integrert i byggesak på KRS kommune nettside
- KRS kommune Innafor
 - Skulle ha vist dem til den siden, den sier egentlig alt om søknadspliktighet
- For dispensasjon så må man søke dispensasjon
 - Hvis man ikke er søknadspliktig, så trenger man bare dispensasjon
 - Om søknadspliktig, så må sende søknad og dispensasjon som blir et vedlegg

Søker notater:

- Har kjøpt hytte
- Lest seg opp på området og hva de kan gjøre
- Innenfor 50 meters beltet
- Søknad som ble innvilget i 2021, men har gått ut
- Tidligere ble søkt om mye om det bygget

- Ikke rive, men vil rehabilitere hytten

- Hva er søknadsplikting og hva må man søke dispensasjon på
- Spørsmål: rive av peis, søknadspliktig eller dispensasjon?
- Konklusjon: Søknadspliktig, ikke dispensasjon, rehabilitering behandles som fasadeendring
- Skifte vindu går fint
- Rive peis og pipe innvendig og utvendig, legge ny pipe

- Lekker i taket og uisolert
 - Vil legge på 10 cm isolasjon
 - Vil forhøye taket med 15 cm

- Søknadspliktig? Ja
- Dispensasjonspliktig? Nei

- Vil ta taket ut 1 meter, ikke utvide hytta, bare taket
- Dispensasjonspliktig? Ja
- Søknadspliktig? Nei
 - I seg selv, så er ikke tiltaket søknadspliktig, man avklarer dispensasjonen først og så melder man inn en melding om tiltaket når man er ferdig. Kan ikke sende inn tiltaket før
 - Det er mye som er unntatt søknadspliktig

- Vil legge og utvide platting
 - Dispensasjonspliktig? Ja
 - Søknadspliktig? Ja

- Lurer på veldig spesifikke ting til hytta
 - Vil endre veldig spesifikke ting samtidig

- Gebyr
 - Det blir gebyr for tiltakene
 - Vil vite spesifikk gebyr

- Har fått godkjent planer av annekset (separat bygg på samme eiendom)
 - Vil forhøye taket med noen cm
 - Må søke dispensasjon med de nye tegningene? Ja
 - Krav til tegninger
 - Tegna tegningene selv, er det lov? Ja

- Nabovarsel
 - Kan følge frihåndstekst
 - Bare word, kan bruke copilot, blir godkjent
 - Lurt å snakke med naboene

- Tid
 - Vil klare endringene til April
 - Hvor lang tid tar behandlingen av saken?
 - Pleier å si 8-12 uker, men ligger rundt 12 uker på behandling
 - Vil prioritere taket, men det er opp til søker å prioritere det selv

Konklusjon:

- Søke om alt og med dispensasjon
- Fortsette med tegningene, snakke med naboer

Appendix 12 – User Testing Task Plan

Brukertest #1 – Bygge ny garasje

Bakgrunn:

Du er en huseier som bor i en liten kommune. Du har nylig kjøpt en eiendom og ønsker å bygge en **40m² frittstående garasje** på tomten din. For å gjøre dette må du sende inn en **digital byggesøknad** via kommunens nye nettbaserte system.

Dette er første gang du bruker plattformen. Du er komfortabel med å bruke datamaskin til daglig, men er ikke teknisk ekspert. Du har fått beskjed om at søknaden må inneholde **eiendomsinformasjon, prosjektbeskrivelse, tegninger og situasjonsplan**, og at alt må leveres digitalt via den nye løsningen.

Du har allerede vært i kontakt med en entreprenør og har de grunnleggende dokumentene klare. Nå ønsker du å sende inn søknaden, følge med på fremdriften, og svare på eventuell tilbakemelding fra kommunen.

Oppgaver:

- Opprette en ny søknad for garasjeprojektet.
- Fylle ut prosjektinformasjon, som eiendomsopplysninger, formål og byggets størrelse.
- Bruke det interaktive kartet for å markere planlagt plassering av garasjen på tomten.
- Last opp nødvendige dokumenter, som arkitekttegninger og situasjonsplan.
- Gå gjennom og sende inn hele søknaden til kommunen.
- Sjekke søknadsstatus etter innsending for å se om den er til behandling, eller om det kreves mer informasjon.

Brukertest #2 – Bruksendring

Bakgrunn:

Du er en huseier i en liten kommune. I eneboligen din har du et **20m² stort bodrom i kjelleren**, som du ønsker å gjøre om til et soverom med et lite bad. Denne typen ombygging krever bruksendring, fordi rommet skal brukes som varig oppholdsrom.

Dette er første gang du bruker den digitale byggesøknadsplattformen, og du må sende inn all nødvendig dokumentasjon – inkludert **planløsninger før og etter**, samt informasjon om **ventilasjon, brannsikkerhet og takhøyde**.

Oppgaver:

- Starte en ny søknad og velge riktig søknadstype: "Bruksendring".
- Fylle ut prosjektdetaljer, som formål med endringen og hvilket areal som omfattes.
- Laste opp plantegninger som viser hvordan boden skal bli et godkjent oppholdsrom.
- Legge ved dokumentasjon på ventilasjon, lysinnslipp, takhøyde og rømningsveier.
- Bruke kart- eller plantegningsverktøy for å markere hvor i bygningen endringen skjer.
- Send inn søknaden og sjekke at alle obligatoriske felt er fylt ut.
- Følg med på statusen etter innsending.

Brukertest #3 – Riving

Bakgrunn:

Du er ny huseier og har nylig kjøpt et eldre hus. I hagen står det et gammelt, ustabilt uthus i tre som du ønsker å rive, da det ikke lenger er trygt å bruke. I henhold til kommunale regler må du sende inn en **rivesøknad** for å varsle kommunen før bygget fjernes.

Du må inkludere et **bilde av bygget**, og forklare hvorfor det skal fjernes.

Oppgaver:

- Opprette en ny søknad og velge "Rive" som tiltakstype.
- Legge inn informasjon om bygget som skal rives (størrelse, materialer, tilstand).
- Laste opp bilder av uthuset og vise dets plassering på eiendommen.
- Bruk kartverktøy for å markere byggets plassering.
- Send inn søknaden og få bekreftelse.

Brukertest #4 – Chatbot

Bakgrunn:

Du holder på å fylle ut en digital byggesøknad for å bygge et **30m² tilbygg** til huset ditt. Du er usikker på om du må **varsle naboene** som en del av prosessen. Du er i ferd med å fylle ut “Prosjektbeskrivelse”-delen og ser et felt som handler om “Nabovarsel”, men det er ikke klart for deg om det gjelder ditt tilfelle.

I stedet for å ringe kommunen, velger du å bruke den innebygde **chatbotten** for å få hjelp.

Oppgaver:

- Starte chatbotten ved å klikke på chatte-ikonet på skjermen.
- Skrive inn spørsmålet ditt: “Må jeg sende nabovarsel for et 30m² tilbygg på huset mitt?”
- Motta svar som forklarer reglene for når nabovarsel kreves, gjerne basert på bygningstype, størrelse, avstand til nabogrense og reguleringsplan.
- Følge en veiledningslenke som chatboten gir, til offisiell informasjon om kravene.
- Stille et oppfølgingsspørsmål, f.eks. “Hva hvis det er under 4 meter fra nabogrensen?”
- Motta et klart og forståelig svar.

Appendix 13 – Sprint Retrospectives

Throughout the 17 sprint retrospectives, several recurring themes and takeaways emerged that reflect the group's ongoing efforts to improve collaboration, workflow, and overall project execution:

- **Consistent Strengths:**
The group maintained strong internal routines, a collaborative work dynamic, and steady progress across all development phases. Regular communication and alignment with the stakeholder contributed to effective planning and delivery of sprint goals.
- **Efficiency and Time Use:**
A common improvement point was the need to work more efficiently, particularly during self-study periods. While overall productivity was high, several sprints highlighted challenges in maintaining consistent momentum and making optimal use of available time.
- **Balance Between Focus and Rest:**
Multiple retrospectives emphasized the importance of balancing efficiency with sustainable work habits. The group frequently reminded themselves to take breaks to avoid fatigue, especially during development-intensive phases.
- **Communication and Oversight:**
Clear communication within the group was occasionally identified as an area for improvement. Ensuring alignment on ongoing tasks and having better oversight of individual progress were recurring topics, particularly in sprints with more distributed work.
- **Stakeholder Integration and Feedback:**
Engaging with the stakeholder through regular meetings and steering group sessions proved to be a valuable source of direction and motivation. Feedback from these sessions was continuously incorporated into sprint planning and product refinement.

These themes shaped a reflective and responsive development process, where continuous learning and iterative improvement were central to the group's agile methodology.

Appendix 14 – Meeting Minutes (møtereferat)

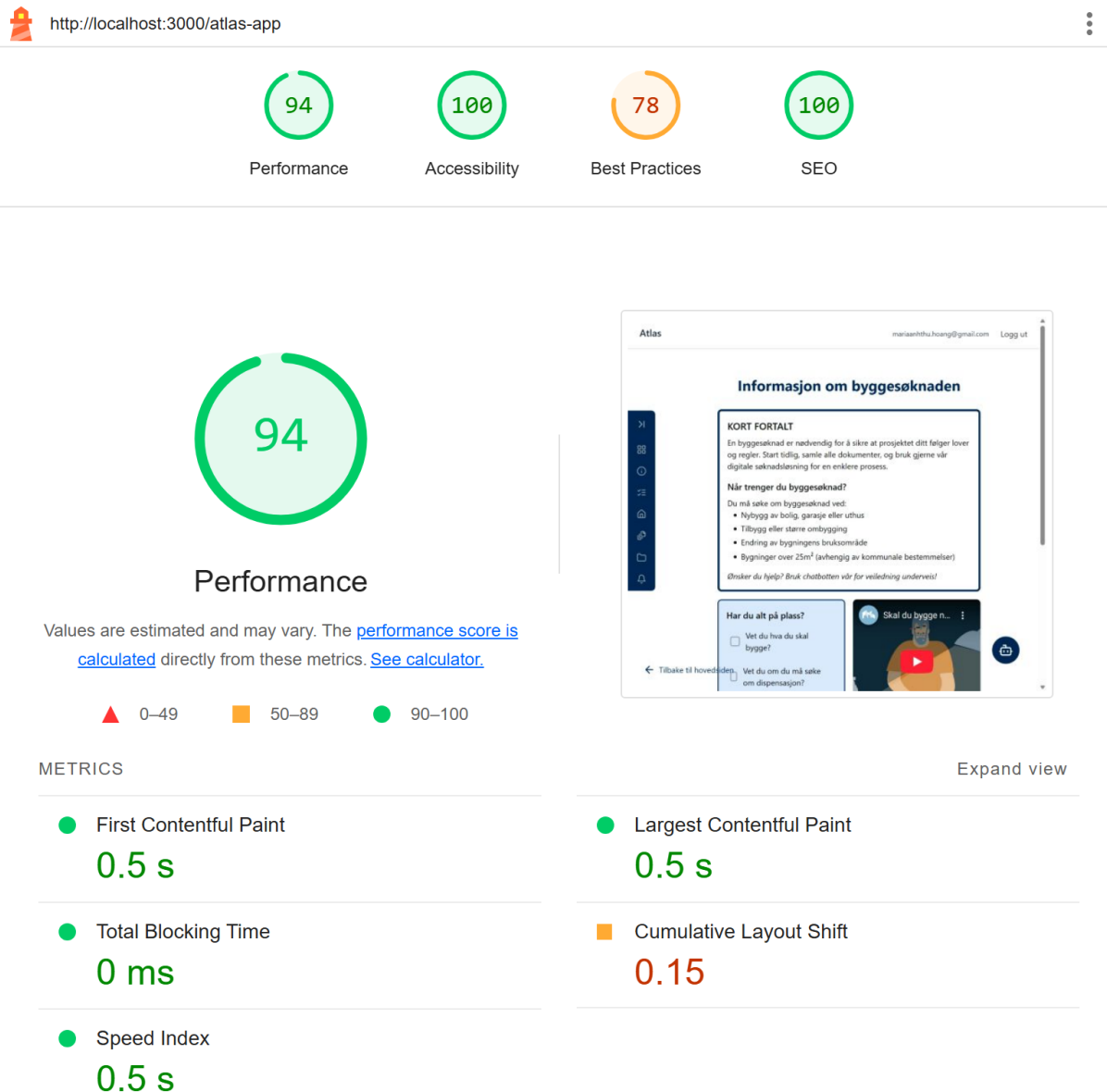
Link to the Meeting Minutes (møtereferat) from the steering group meetings (styringsgruppemøtene): https://drive.google.com/drive/folders/1BJ_xbZNX4E2Cwrc-QM81uSN5DSAuBcEZ?usp=sharing

Appendix 15 – Presentation Materials (Status 1 & Status 2)

Link to the presentations from Status 1 and Status 2:

https://drive.google.com/drive/folders/1cKIAytjHDxkn8-g074LtT9WjNMEZVbfS?usp=drive_link

Appendix 16 - Lighthouse Report



 View Treemap



Show audits relevant to: [All](#) [FCP](#) [LCP](#) [TBT](#) [CLS](#)

DIAGNOSTICS

- ▲ Avoid large layout shifts — 8 layout shifts found
- ▲ Reduce unused JavaScript — Potential savings of 81 KiB
- ▲ Page prevented back/forward cache restoration — 2 failure reasons
- Serve static assets with an efficient cache policy — 6 resources found
- Some third-party resources can be lazy loaded with a facade — 1 facade alternative available
- Eliminate render-blocking resources — Potential savings of 0 ms
- Avoid serving legacy JavaScript to modern browsers — Potential savings of 11 KiB
- Avoid non-composited animations — 10 animated elements found
- Avoid chaining critical requests — 3 chains found
- Minimize third-party usage — Third-party code blocked the main thread for 0 ms

○ Largest Contentful Paint element — 530 ms



More information about the performance of your application. These numbers don't [directly affect](#) the Performance score.

PASSED AUDITS (27)

Show



Accessibility

These checks highlight opportunities to [improve the accessibility of your web app](#). Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so [manual testing](#) is also encouraged.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

Hide

○ Interactive controls are keyboard focusable



☐ Interactive elements indicate their purpose and state	▼
☐ The page has a logical tab order	▼
☐ Visual order on the page follows DOM order	▼
☐ User focus is not accidentally trapped in a region	▼
☐ The user's focus is directed to new content added to the page	▼
☐ HTML5 landmark elements are used to improve navigation	▼
☐ Offscreen content is hidden from assistive technology	▼
☐ Custom controls have associated labels	▼
☐ Custom controls have ARIA roles	▼

These items address areas which an automated testing tool cannot cover. Learn more in our guide on [conducting an accessibility review](#).

PASSED AUDITS (22)

Show

NOT APPLICABLE (35)

Show



Best Practices

GENERAL

▲ Uses third-party cookies — 30 cookies found	▼
▲ Issues were logged in the Issues panel in Chrome Devtools	▼
○ Detected JavaScript libraries	▼

TRUST AND SAFETY

○ Ensure CSP is effective against XSS attacks	▼
○ Use a strong HSTS policy	▼
○ Ensure proper origin isolation with COOP	▼
○ Mitigate clickjacking with XFO or CSP	▼

PASSED AUDITS (12)

Show

NOT APPLICABLE (2)

Show



SEO

These checks ensure that your page is following basic search engine optimization advice. There are many additional factors Lighthouse does not score here that may affect your search ranking, including performance on [Core Web Vitals](#). [Learn more about Google Search Essentials](#).

ADDITIONAL ITEMS TO MANUALLY CHECK (1)

Hide

☐ Structured data is valid



Run these additional validators on your site to check additional SEO best practices.

PASSED AUDITS (7)

Show

NOT APPLICABLE (3)

Show

Captured at May 10, 2025, 3:02 PM
GMT+2

Emulated Desktop with Lighthouse
12.5.1

Single page session

Initial page load

Custom throttling

Using Chromium 136.0.0.0 with
devtools

Generated by **Lighthouse** 12.5.1 | [File an issue](#)